Introduction to Linear Programming

BUS 104: Decision Analysis and Management Science

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October 12, 2026

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Agenda: Linear Programming Applications

Welcome! In this presentation, we will explore the power of Linear Programming (LP) through a series of foundational business and operational problems.

We will cover how to formulate and solve:

1. **Product Mix Problem:** Maximizing manufacturing profit (Par, Inc.)
2. **Blending Problems (I & II):** Mixing ingredients to meet specifications at lowest cost (e.g., Coffee Blending)
3. **Work Scheduling Problem:** Assigning labor shifts to meet demand
4. **Transportation Problem:** Minimizing shipping costs from origins to destinations
5. **Assignment Problem:** Matching tasks to individuals optimally (e.g., Relay Teams)
6. **Shortest Path Problem:** Finding the most efficient route through a network

Our Approach

For each problem category, we will follow a rigorous 3-step methodology:

- **Step 1: The Business Problem.** Understanding the scenario, the physical constraints, and the managerial objective.
- **Step 2: Mathematical Formulation.** Translating English into Algebra (defining Decision Variables, Objective Function, and Constraints).
- **Step 3: Python Implementation.** Using the `PuLP` library to computationally solve the model and extract business insights.

Let's begin with our first category: The Product Mix Problem.

Product Mix Example: Par, Inc.

Goal: Maximize weekly profit by deciding how many standard and deluxe golf bags to produce.

The Data:

Department	**Standard Bag**	**Deluxe Bag**	**Hours Available**
Cutting & Dyeing	7/10 hr	1 hr	630 hrs
Sewing	1/2 hr	5/6 hr	600 hrs
Finishing	1 hr	2/3 hr	708 hrs
Inspection & Pack.	0.1 hr	1/4 hr	135 hrs
Profit/Bag	**$10**	**$9**	

- *Assumption:* Demand is unlimited; we can sell everything we make.
- **Question:** Formulate a mathematical model to find the optimal production plan.

Par, Inc. Golf Bag Production Problem

Decision variables.

Let x_s = number of standard golf bags produced per week,
and x_d = number of deluxe golf bags produced per week.

Both $x_s, x_d \geq 0$.

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Objective function.
Maximize weekly profit:

$$\max \ Z = 10x_s + 9x_d.$$

Subject to capacity constraints:

$$\frac{7}{10}x_s + 1x_d \leq 630 \qquad \text{(Cutting \& Dyeing)}$$

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Subject to capacity constraints:

$$\frac{7}{10}x_s + 1x_d \leq 630 \qquad \text{(Cutting \& Dyeing)}$$
$$\frac{1}{2}x_s + \frac{5}{6}x_d \leq 600 \qquad \text{(Sewing)}$$

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Decision variables.
Let x_s = number of standard golf bags produced per week,
and x_d = number of deluxe golf bags produced per week.

Both $x_s, x_d \geq 0$.

Objective function.
Maximize weekly profit:

$$\max \ Z = 10x_s + 9x_d.$$

Subject to capacity constraints:

$$\begin{aligned} \tfrac{7}{10}x_s + 1x_d &\leq 630 && \text{(Cutting \& Dyeing)} \\ \tfrac{1}{2}x_s + \tfrac{5}{6}x_d &\leq 600 && \text{(Sewing)} \\ 1x_s + \tfrac{2}{3}x_d &\leq 708 && \text{(Finishing)} \end{aligned}$$

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Decision variables.
Let x_s = number of standard golf bags produced per week,
and x_d = number of deluxe golf bags produced per week.

Both $x_s, x_d \geq 0$.

Objective function.
Maximize weekly profit:

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Subject to capacity constraints:

$$\begin{aligned}
\tfrac{7}{10}x_s + 1x_d &\leq 630 && \text{(Cutting \& Dyeing)} \\
\tfrac{1}{2}x_s + \tfrac{5}{6}x_d &\leq 600 && \text{(Sewing)} \\
1x_s + \tfrac{2}{3}x_d &\leq 708 && \text{(Finishing)} \\
0.1x_s + \tfrac{1}{4}x_d &\leq 135 && \text{(Inspection \& Packaging)} \\
x_s, x_d &\geq 0.
\end{aligned}$$

Par, Inc. Golf Bag Production Solution (PuLP)

```
import pulp as pl

# Create the problem
prob = pl.LpProblem("Par_Inc_Golf_Bags", pl.LpMaximize)

# Define variables directly into the model
x_s = prob.add_variable("Standard_Bags", lowBound=0)
x_d = prob.add_variable("Deluxe_Bags", lowBound=0)

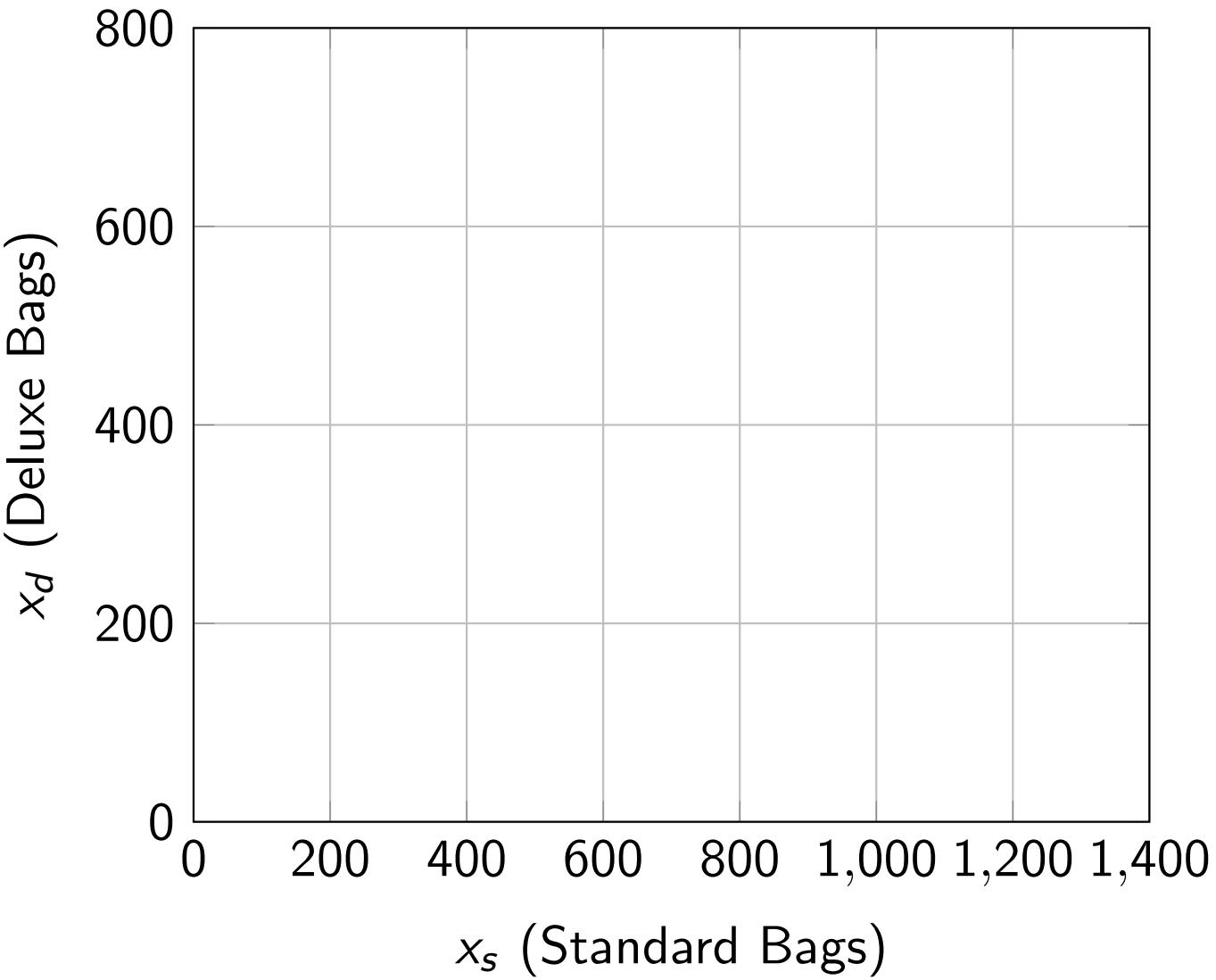
# Objective function
prob += 10 * x_s + 9 * x_d, "Maximize_Profit"

# Constraints
prob += 0.7 * x_s + 1 * x_d <= 630, "Cutting_Dyeing"
prob += 0.5 * x_s + 5/6 * x_d <= 600, "Sewing"
prob += 1 * x_s + 2/3 * x_d <= 708, "Finishing"
prob += 0.1 * x_s + 0.25 * x_d <= 135, "Inspection_Packaging"

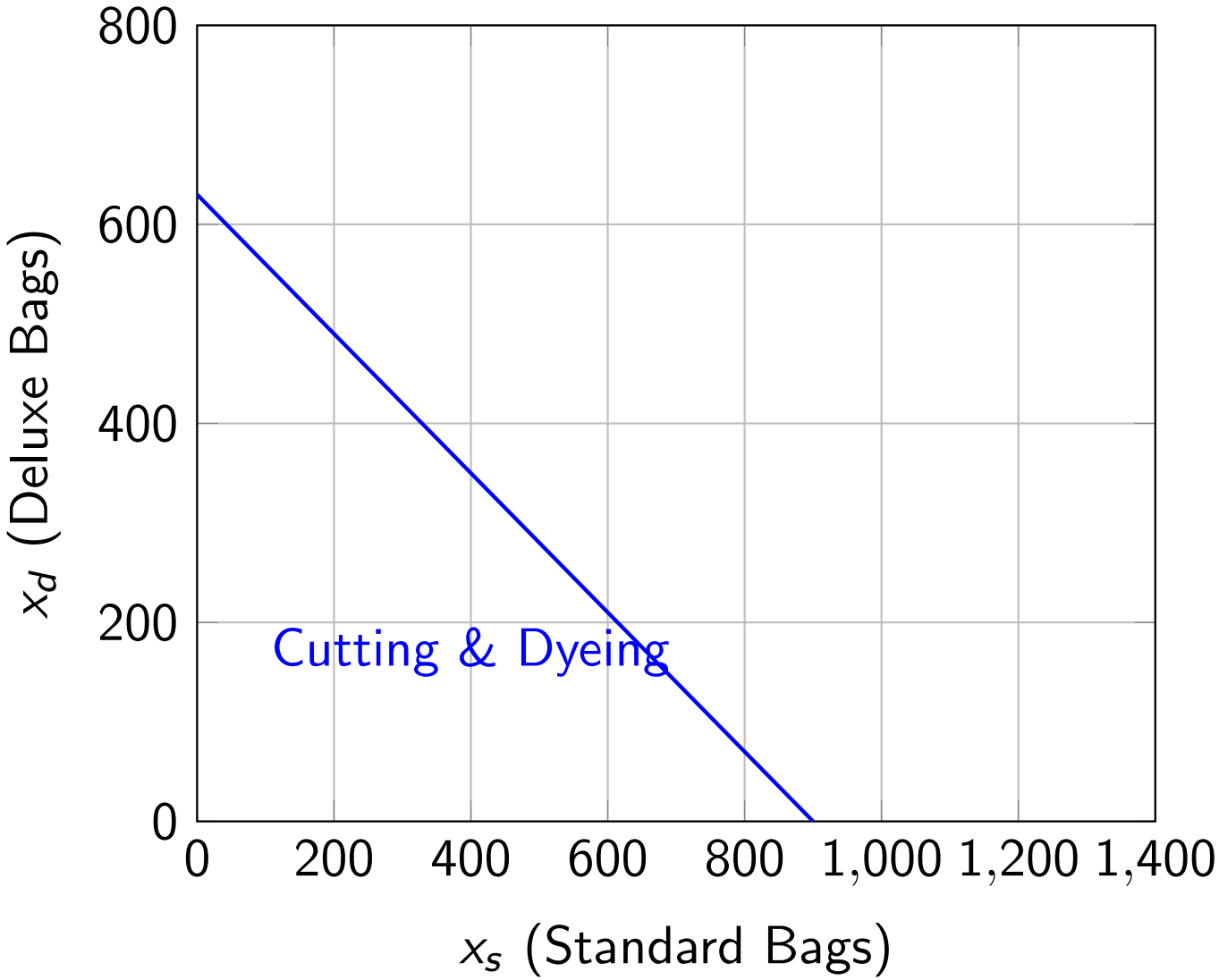
# Solve the problem
res = prob.solve()

# Print results
print("Status:", res.status.name)
print("Optimal Solution:")
print(f"Standard Bags = {x_s.varValue}")
print(f"Deluxe Bags = {x_d.varValue}")
print(f"Maximum Profit = ${pl.value(prob.objective)}")
```

Graphical Solution – Step 1: Set up the Axes



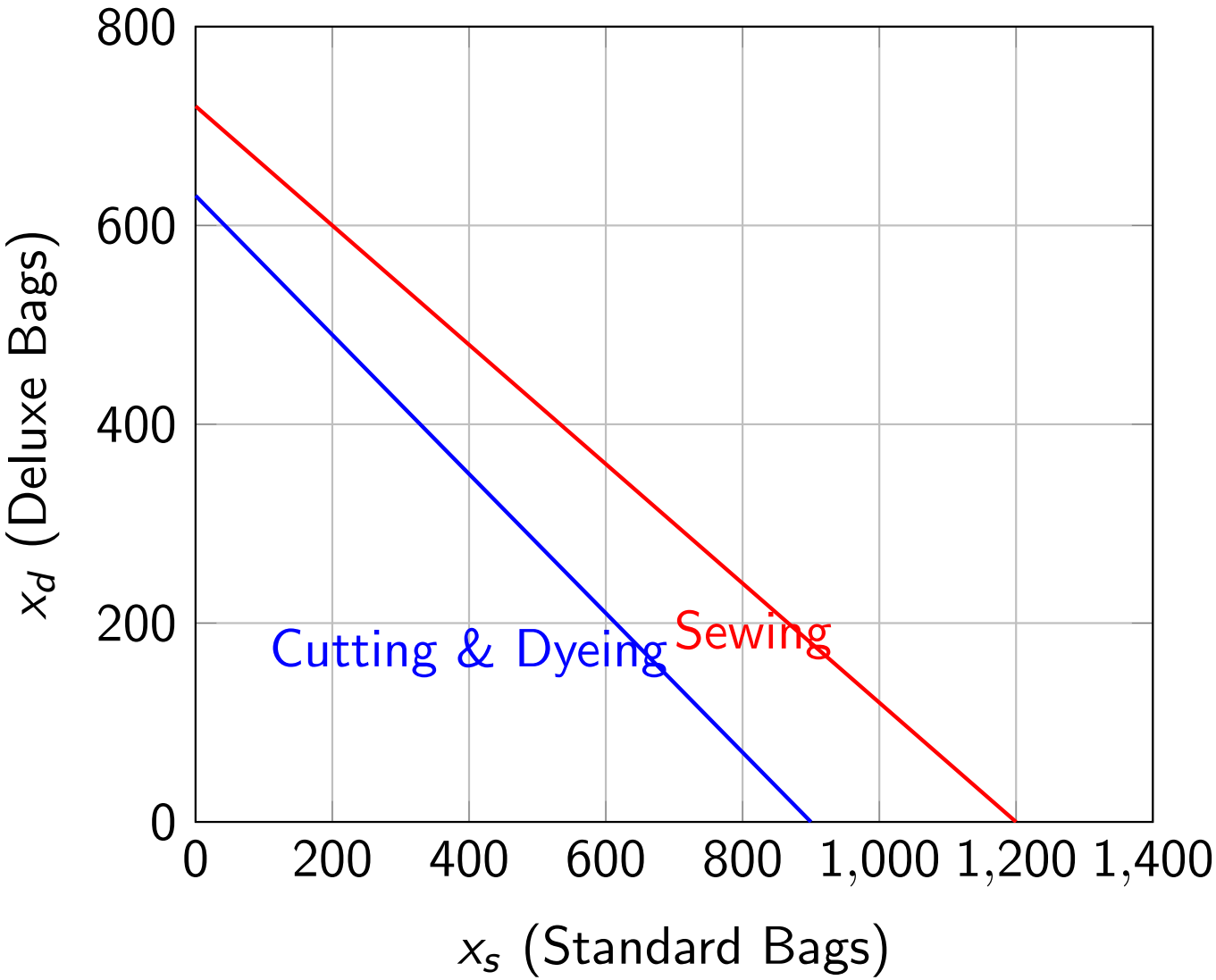
Graphical Solution – Step 2: Add C&D Constraint



$0.7x_s + x_d \leq 630$

Feasible region is below the line.

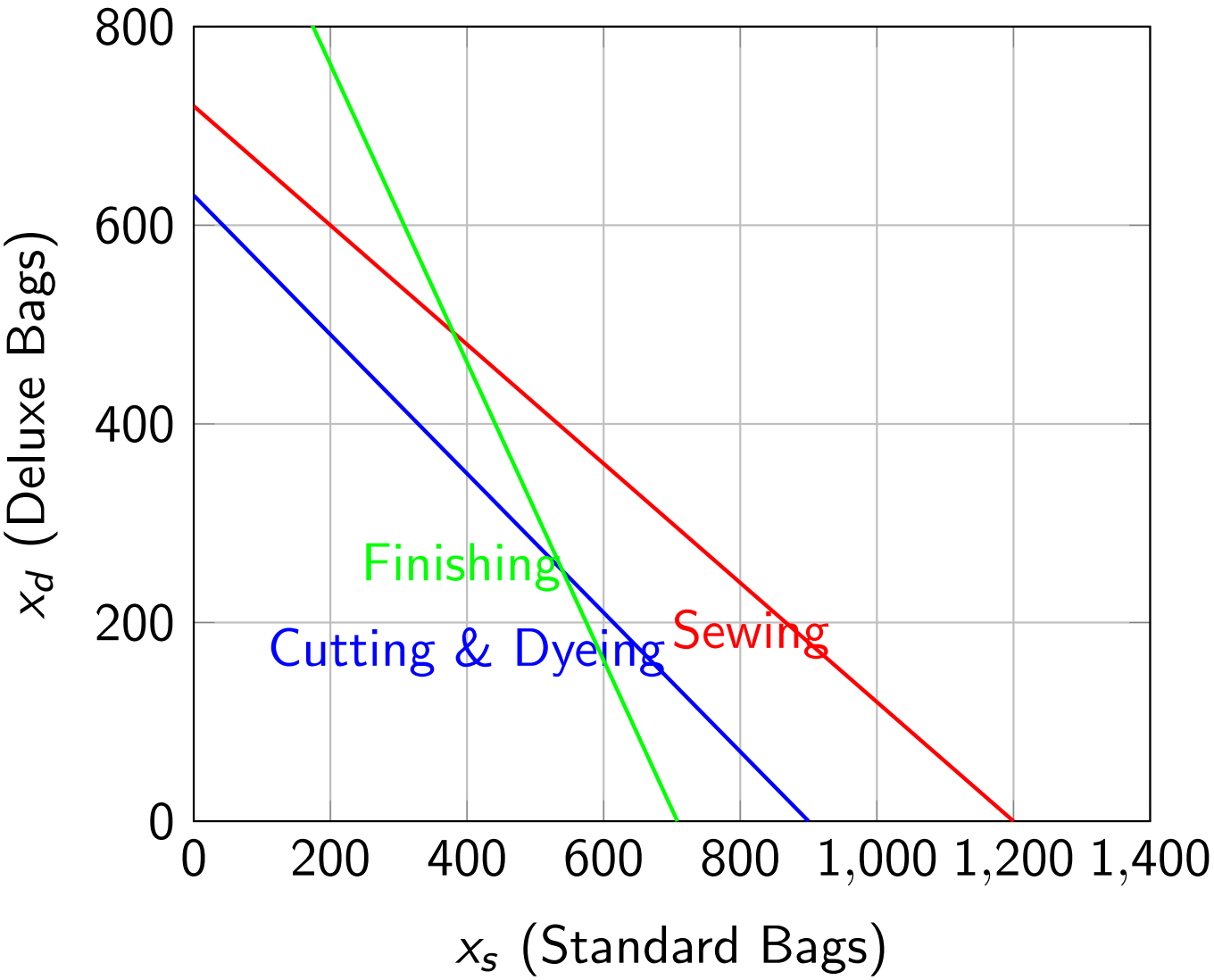
Graphical Solution – Step 3: Add Sewing Constraint



$0.5x_s + \frac{5}{6}x_d \leq 600$

Feasible region is below the line.

Graphical Solution – Step 4: Add Finishing Constraint

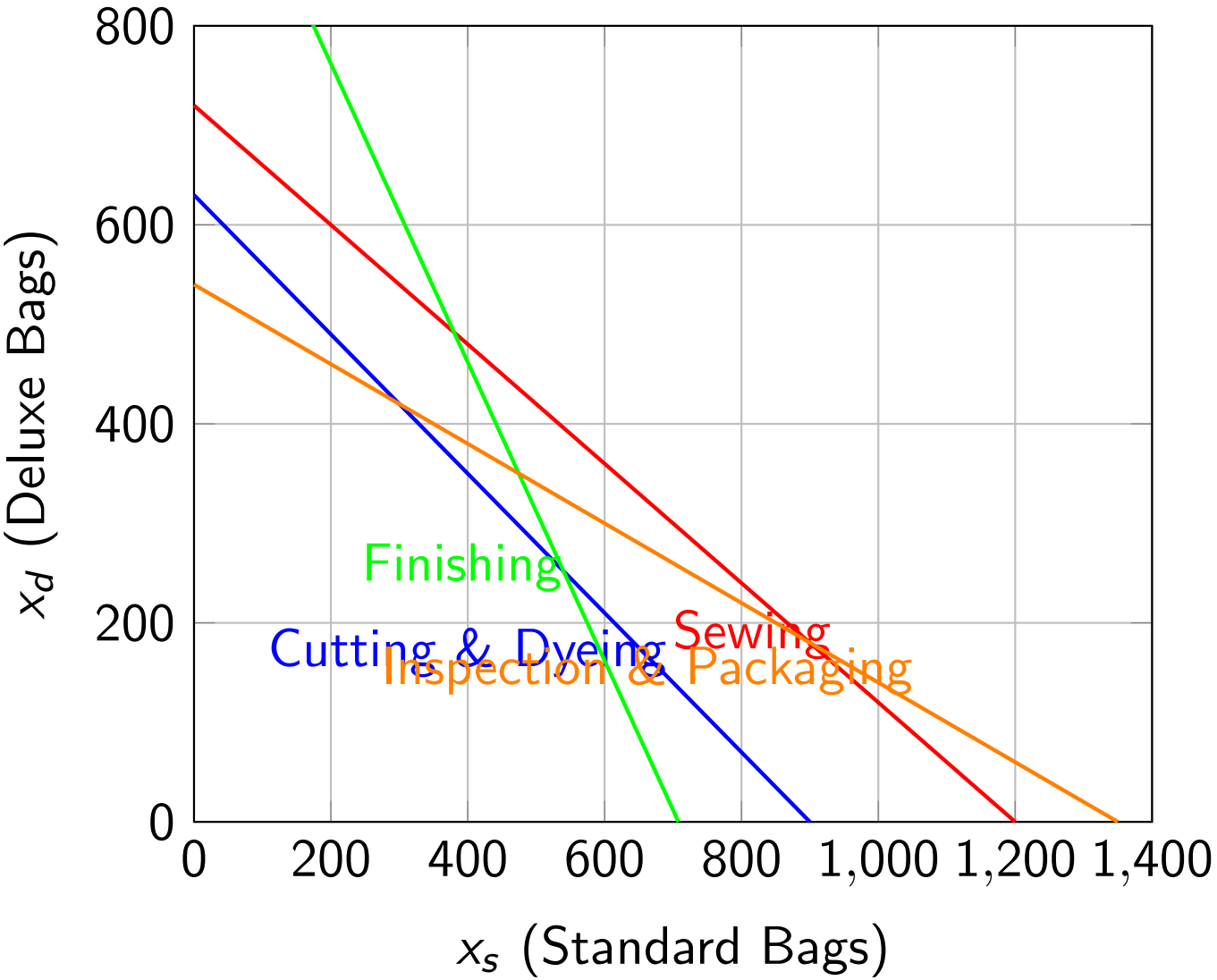


$x_s + \frac{2}{3}x_d \leq 708$

Feasible region is below the line.

Graphical Solution – Step 5: Add Inspection Constraint

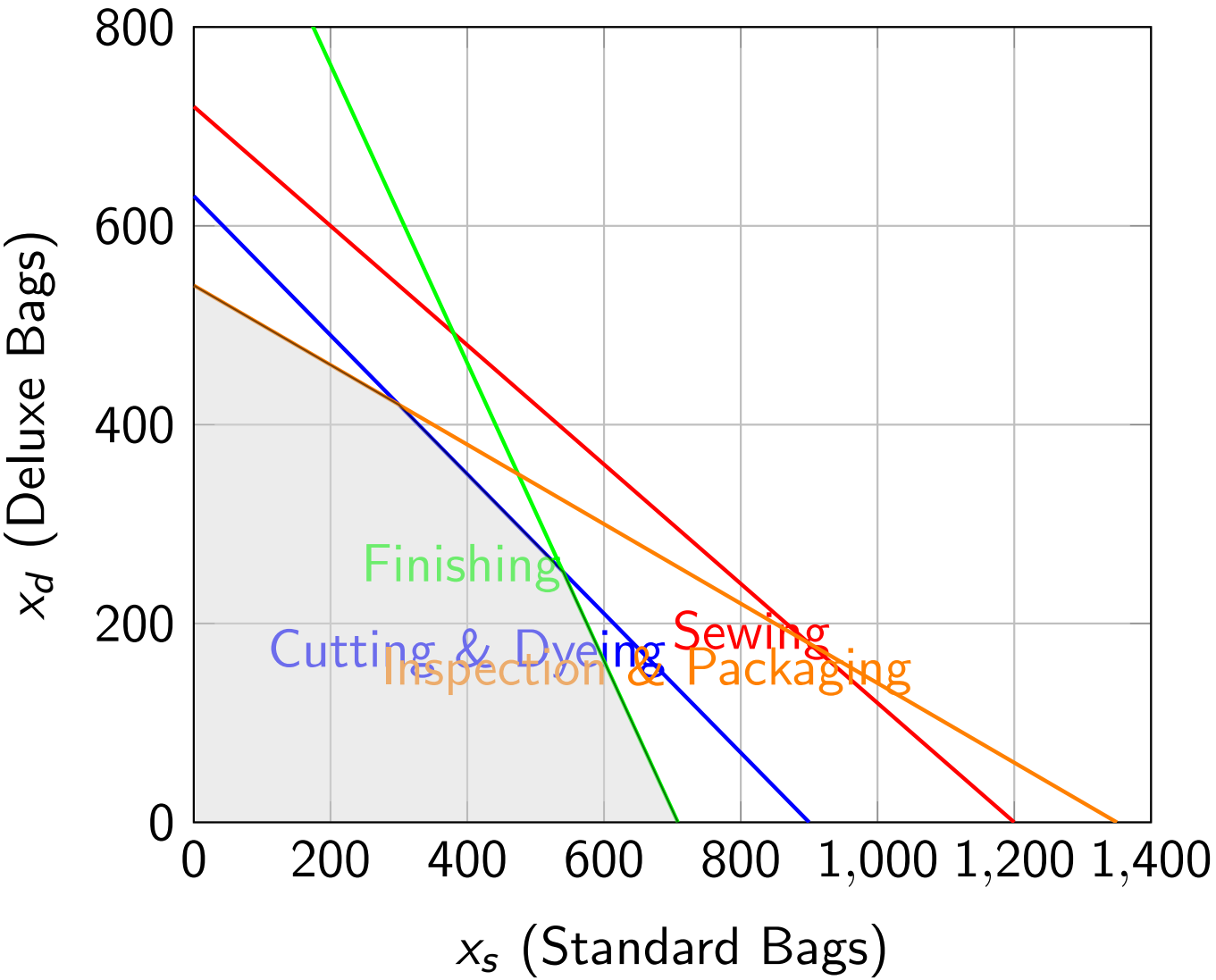
$0.1x_s + 0.25x_d \leq 135$



Feasible region is below the line.

Graphical Solution – Step 6: Feasible Region

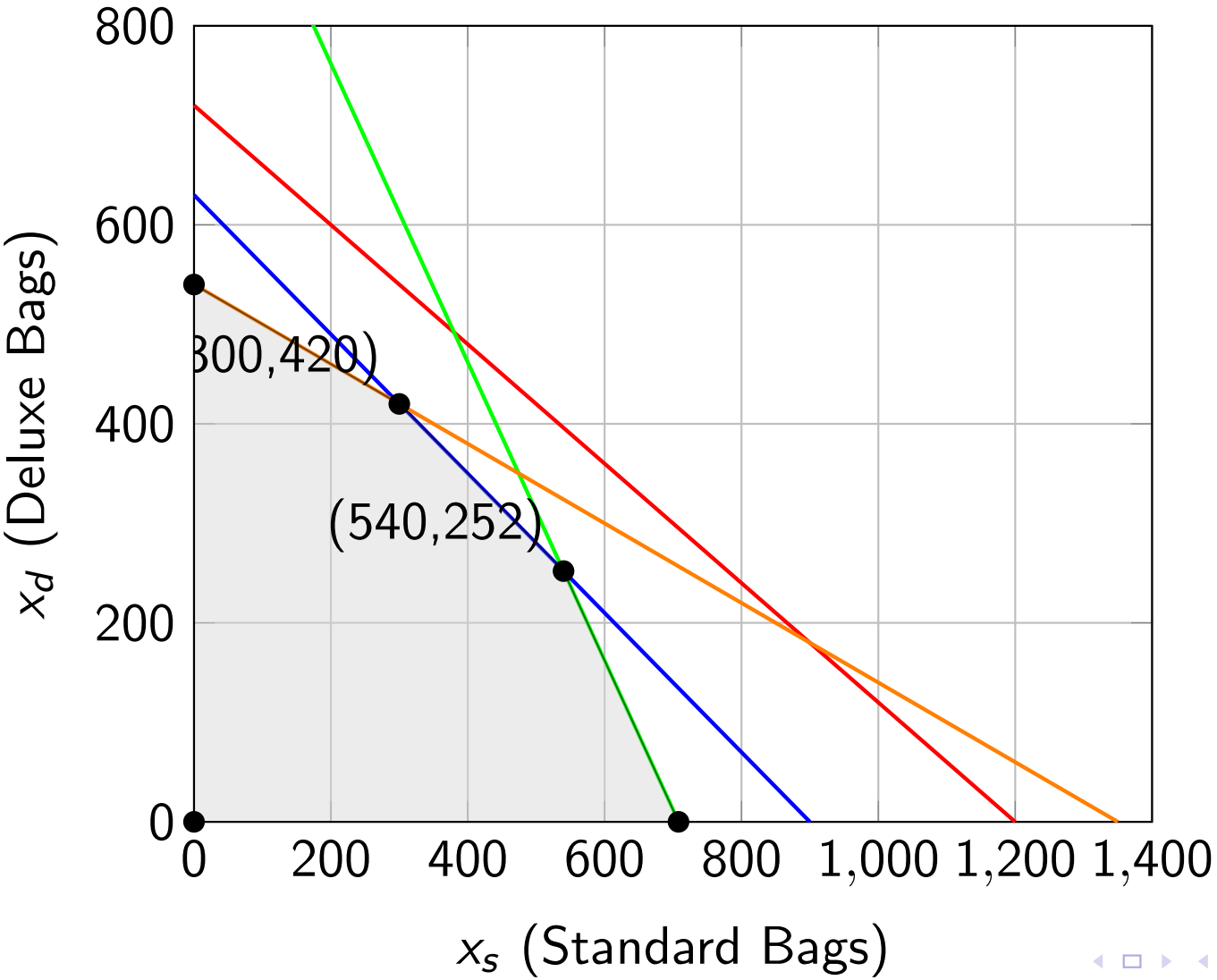
The feasible region is the area below all constraint lines and above the



axes.

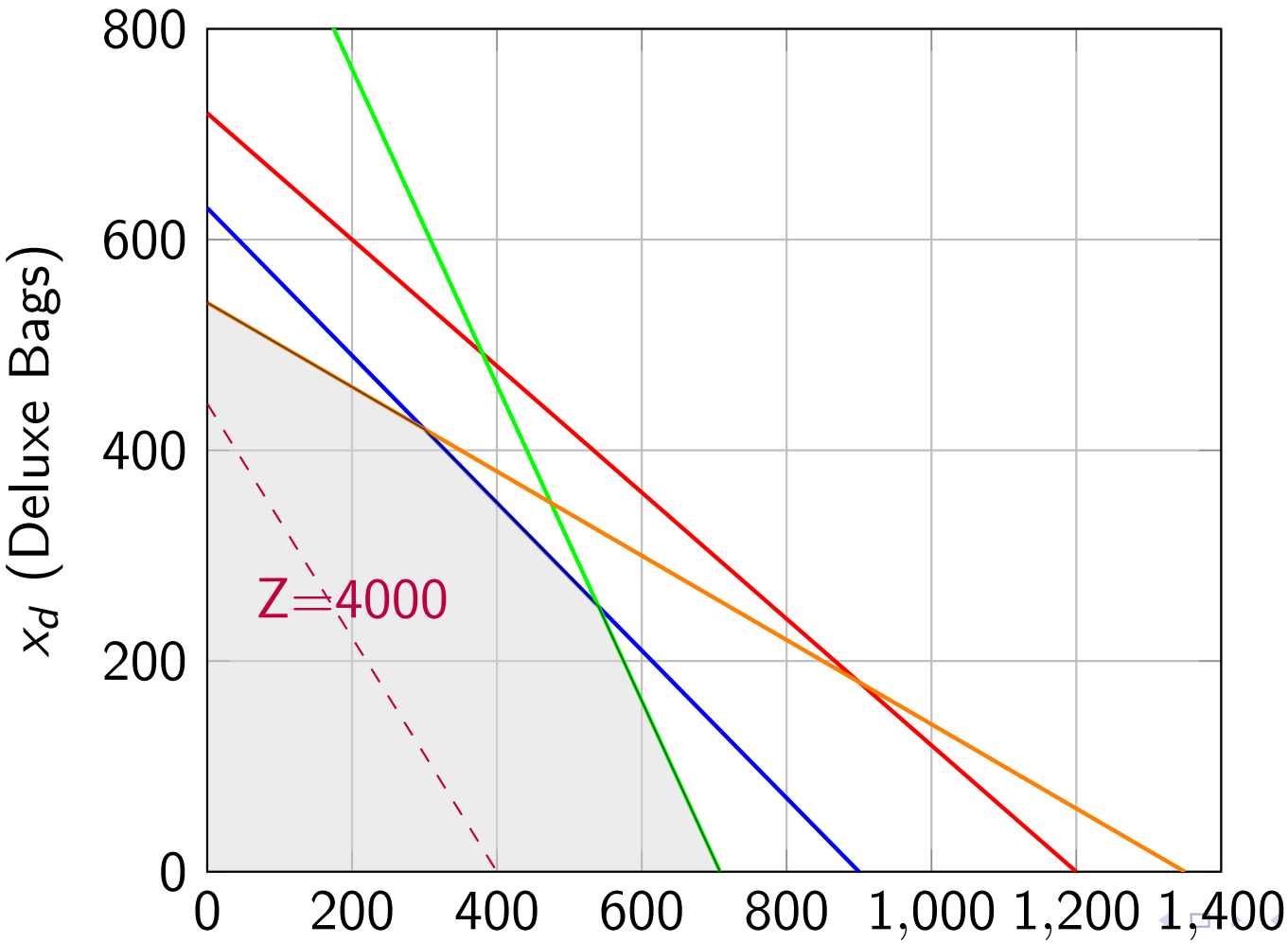
Graphical Solution – Step 7: Corner Points

Identify the corner points of the feasible region: - (0, 0) - (0, 540) - (300, 420) - (540, 252) - (708, 0)



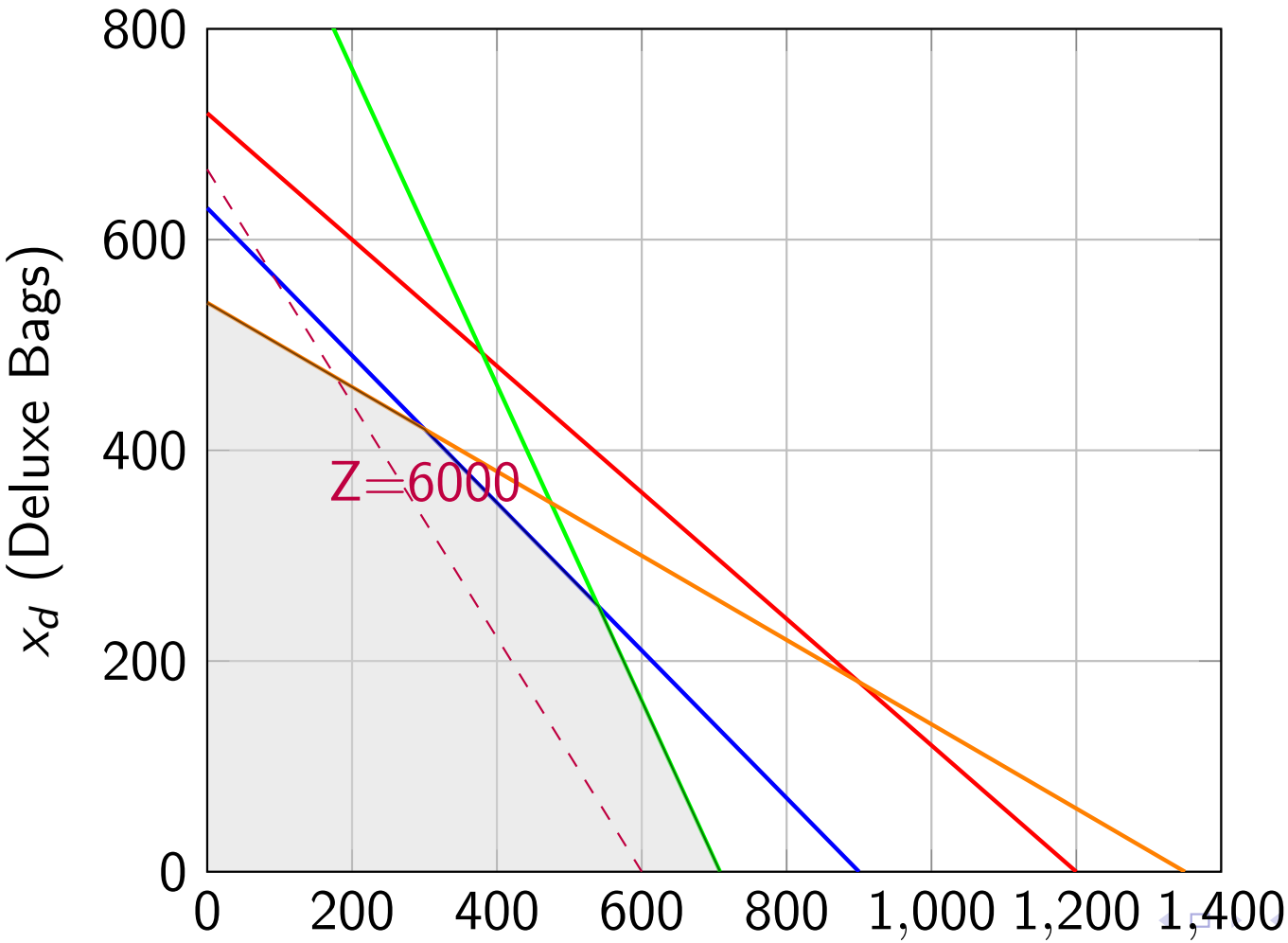
Graphical Solution – Step 8: Objective Function at Low Profit

Objective: $Z = 10x_s + 9x_d = 4000$ Line:
$x_d = \frac{4000}{9} - \frac{10}{9}x_s \approx 444.44 - 1.111x_s$



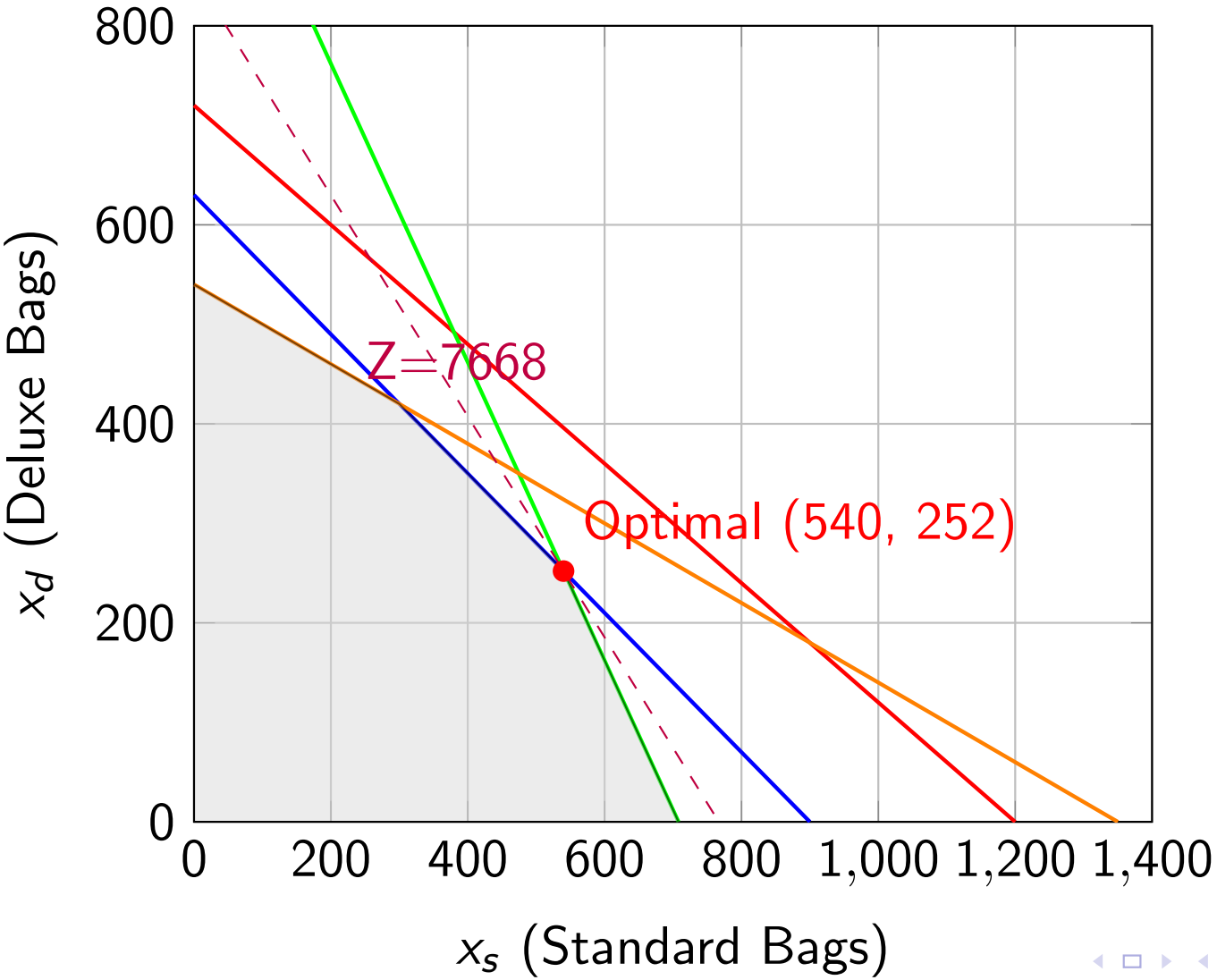
Graphical Solution – Step 9: Objective Function at Higher Profit

Objective: $Z = 10x_s + 9x_d = 6000$ Line:
$x_d = \frac{6000}{9} - \frac{10}{9}x_s \approx 666.67 - 1.111x_s$



Graphical Solution – Step 10: Optimal Objective Function

Objective: $Z = 10x_s + 9x_d = 7668$ Line: $x_d = \frac{7668}{9} - \frac{10}{9}x_s \approx 852 - 1.111x_s$
The line touches the feasible region at the optimal point (540, 252).



Blending Example: MediBlend

Goal: Find the cheapest way to produce today's batch of MediBlend by mixing four chemicals.

The Rules & Constraints:

- **Batch Size:** Must produce *at least* 1,000 lbs total.
- **Ingredient Minimums:** The final mix must contain at least 8% of A, 4% of B, and 2% of C (by weight).
- **Supply Constraint:** Must use *at least* 100 lbs of Chemical 2.

The Data:

Chemical	Cost/lb	Ingredient A	Ingredient B	Ingredient C
Chemical 1	\$8	0.03	0.02	0.01
Chemical 2	\$10	0.06	0.04	0.01
Chemical 3	\$11	0.10	0.03	0.04
Chemical 4	\$14	0.12	0.09	0.04

Blending Example: LP Formulation

Let x_1, x_2, x_3, x_4 denote the pounds of chemicals (1,2,3,4) used to produce today's batch. The objective is to minimize total cost subject to meeting the minimum batch size, the ingredient percentage requirements, the minimum usage of chemical 2, and nonnegativity:

Blending Example: LP Formulation

Let x_1, x_2, x_3, x_4 denote the pounds of chemicals (1,2,3,4) used to produce today's batch. The objective is to minimize total cost subject to meeting the minimum batch size, the ingredient percentage requirements, the minimum usage of chemical 2, and nonnegativity:

$$\min_{x_1,x_2,x_3,x_4} \quad 8x_1 + 10x_2 + 11x_3 + 14x_4$$

Blending Example: LP Formulation

Let x_1, x_2, x_3, x_4 denote the pounds of chemicals (1,2,3,4) used to produce today's batch. The objective is to minimize total cost subject to meeting the minimum batch size, the ingredient percentage requirements, the minimum usage of chemical 2, and nonnegativity:

$$\begin{aligned} \min_{x_1,x_2,x_3,x_4} \quad & 8x_1 + 10x_2 + 11x_3 + 14x_4 \\ \text{s.t.} \quad & x_1 + x_2 + x_3 + x_4 \geq 1000 \end{aligned}$$

Blending Example: LP Formulation

Let x_1, x_2, x_3, x_4 denote the pounds of chemicals (1,2,3,4) used to produce today's batch. The objective is to minimize total cost subject to meeting the minimum batch size, the ingredient percentage requirements, the minimum usage of chemical 2, and nonnegativity:

$$\begin{aligned} \min_{x_1,x_2,x_3,x_4} \quad & 8x_1 + 10x_2 + 11x_3 + 14x_4 \\ \text{s.t.} \quad & x_1 + x_2 + x_3 + x_4 \geq 1000 \\ & 0.03x_1 + 0.06x_2 + 0.10x_3 + 0.12x_4 \geq 0.08(x_1 + x_2 + x_3 + x_4) \end{aligned}$$

Blending Example: LP Formulation

Let x_1, x_2, x_3, x_4 denote the pounds of chemicals (1,2,3,4) used to produce today's batch. The objective is to minimize total cost subject to meeting the minimum batch size, the ingredient percentage requirements, the minimum usage of chemical 2, and nonnegativity:

$$\begin{aligned} \min_{x_1,x_2,x_3,x_4} \quad & 8x_1 + 10x_2 + 11x_3 + 14x_4 \\ \text{s.t.} \quad & x_1 + x_2 + x_3 + x_4 \geq 1000 \\ & 0.03x_1 + 0.06x_2 + 0.10x_3 + 0.12x_4 \geq 0.08(x_1 + x_2 + x_3 + x_4) \\ & 0.02x_1 + 0.04x_2 + 0.03x_3 + 0.09x_4 \geq 0.04(x_1 + x_2 + x_3 + x_4) \end{aligned}$$

Blending Example: LP Formulation

Let x_1, x_2, x_3, x_4 denote the pounds of chemicals (1,2,3,4) used to produce today's batch. The objective is to minimize total cost subject to meeting the minimum batch size, the ingredient percentage requirements, the minimum usage of chemical 2, and nonnegativity:

$$\begin{aligned}
\min_{x_1,x_2,x_3,x_4} \quad & 8x_1 + 10x_2 + 11x_3 + 14x_4 \\
\text{s.t.} \quad & x_1 + x_2 + x_3 + x_4 \geq 1000 \\
& 0.03x_1 + 0.06x_2 + 0.10x_3 + 0.12x_4 \geq 0.08(x_1 + x_2 + x_3 + x_4) \\
& 0.02x_1 + 0.04x_2 + 0.03x_3 + 0.09x_4 \geq 0.04(x_1 + x_2 + x_3 + x_4) \\
& 0.01x_1 + 0.01x_2 + 0.04x_3 + 0.04x_4 \geq 0.02(x_1 + x_2 + x_3 + x_4)
\end{aligned}$$

Blending Example: LP Formulation

Let x_1, x_2, x_3, x_4 denote the pounds of chemicals (1,2,3,4) used to produce today's batch. The objective is to minimize total cost subject to meeting the minimum batch size, the ingredient percentage requirements, the minimum usage of chemical 2, and nonnegativity:

$$\begin{aligned}
\min_{x_1,x_2,x_3,x_4} \quad & 8x_1 + 10x_2 + 11x_3 + 14x_4 \\
\text{s.t.} \quad & x_1 + x_2 + x_3 + x_4 \geq 1000 \\
& 0.03x_1 + 0.06x_2 + 0.10x_3 + 0.12x_4 \geq 0.08(x_1 + x_2 + x_3 + x_4) \\
& 0.02x_1 + 0.04x_2 + 0.03x_3 + 0.09x_4 \geq 0.04(x_1 + x_2 + x_3 + x_4) \\
& 0.01x_1 + 0.01x_2 + 0.04x_3 + 0.04x_4 \geq 0.02(x_1 + x_2 + x_3 + x_4) \\
& x_2 \geq 100
\end{aligned}$$

Blending Example: LP Formulation

Let x_1, x_2, x_3, x_4 denote the pounds of chemicals (1,2,3,4) used to produce today's batch. The objective is to minimize total cost subject to meeting the minimum batch size, the ingredient percentage requirements, the minimum usage of chemical 2, and nonnegativity:

$$\begin{aligned}
\min_{x_1,x_2,x_3,x_4} \quad & 8x_1 + 10x_2 + 11x_3 + 14x_4 \\
\text{s.t.} \quad & x_1 + x_2 + x_3 + x_4 \geq 1000 \\
& 0.03x_1 + 0.06x_2 + 0.10x_3 + 0.12x_4 \geq 0.08(x_1 + x_2 + x_3 + x_4) \\
& 0.02x_1 + 0.04x_2 + 0.03x_3 + 0.09x_4 \geq 0.04(x_1 + x_2 + x_3 + x_4) \\
& 0.01x_1 + 0.01x_2 + 0.04x_3 + 0.04x_4 \geq 0.02(x_1 + x_2 + x_3 + x_4) \\
& x_2 \geq 100 \\
& x_1, x_2, x_3, x_4 \geq 0
\end{aligned}$$

Blending Example: Classical LP Formulation

In classical Linear Programming (and for many solvers), we rearrange our equations so that all variables are on the left side of the inequality, and all constants are on the right side.

By factoring out the variables from the right side of our ingredient constraints, our model simplifies to:

$$\min_{x_1,x_2,x_3,x_4} \quad 8x_1 + 10x_2 + 11x_3 + 14x_4$$

$$\begin{aligned}
\text{s.t.} \quad & x_1 + x_2 + x_3 + x_4 \geq 1000 && \text{(Batch Size)} \\
& -0.05x_1 - 0.02x_2 + 0.02x_3 + 0.04x_4 \geq 0 && \text{(Ingredient A)} \\
& -0.02x_1 - 0.01x_3 + 0.05x_4 \geq 0 && \text{(Ingredient B)} \\
& -0.01x_1 - 0.01x_2 + 0.02x_3 + 0.02x_4 \geq 0 && \text{(Ingredient C)} \\
& x_2 \geq 100 && \text{(Chemical 2 Min)} \\
& x_1, x_2, x_3, x_4 \geq 0
\end{aligned}$$

Blending Example Solution (PuLP)

```
import pulp as pl

# Create the problem
prob = pl.LpProblem("MediBlend_Blending", pl.LpMinimize)

# Define variables directly into the model
x1 = prob.add_variable("Chemical_1", lowBound=0)
x2 = prob.add_variable("Chemical_2", lowBound=0)
x3 = prob.add_variable("Chemical_3", lowBound=0)
x4 = prob.add_variable("Chemical_4", lowBound=0)

# Objective function
prob += 8 * x1 + 10 * x2 + 11 * x3 + 14 * x4, "Minimize_Cost"

# Constraints
prob += x1 + x2 + x3 + x4 >= 1000, "Batch_Size"
prob += 0.03 * x1 + 0.06 * x2 + 0.10 * x3 + 0.12 * x4 >= 0.08 * (x1 + x2 + x3 + x4), "
     Ingredient_A"
prob += 0.02 * x1 + 0.04 * x2 + 0.03 * x3 + 0.09 * x4 >= 0.04 * (x1 + x2 + x3 + x4), "
     Ingredient_B"
prob += 0.01 * x1 + 0.01 * x2 + 0.04 * x3 + 0.04 * x4 >= 0.02 * (x1 + x2 + x3 + x4), "
     Ingredient_C"
prob += x2 >= 100, "Chemical_2_Min"

# Solve the problem
res = prob.solve()

# Print results
print("Status:", res.status.name)
print("Optimal Solution:")
print(f"Chemical 1 = {x1.varValue} lbs")
print(f"Chemical 2 = {x2.varValue} lbs")
print(f"Chemical 3 = {x3.varValue} lbs")
print(f"Chemical 4 = {x4.varValue} lbs")
print(f"Minimum Cost = ${pl.value(prob.objective)}")
```

Work Scheduling Example

A post office requires different numbers of full-time employees on different days of the week. The number of full-time employees needed on each day is as follows:

Day	Number of Employees
Monday	7
Tuesday	3
Wednesday	5
Thursday	9
Friday	4
Saturday	6
Sunday	1

Table: Weekly Employee Requirements

Union rules state that each full-time employee must work five consecutive days and then receive two days off. For example, an employee who works Monday to Friday must be off Saturday and Sunday. The post office wants to meet its daily requirements using only full-time employees. Formulate an LP that the post office can use to minimize the number of full-time employees that must be hired.

Work Scheduling: Shift Coverage Grid

To formulate the constraints, it helps to visualize which days each shift covers. Employees work 5 consecutive days and then get 2 days off.

Shift Starts	**Mon**	**Tue**	**Wed**	**Thu**	**Fri**	**Sat**	**Sun**
Monday (x_{Mon})	W	W	W	W	W	Off	Off
Tuesday (x_{Tue})	Off	W	W	W	W	W	Off
Wednesday (x_{Wed})	Off	Off	W	W	W	W	W
Thursday (x_{Thu})	W	Off	Off	W	W	W	W
Friday (x_{Fri})	W	W	Off	Off	W	W	W
Saturday (x_{Sat})	W	W	W	Off	Off	W	W
Sunday (x_{Sun})	W	W	W	W	Off	Off	W
Required	**7**	**3**	**5**	**9**	**4**	**6**	**1**

Work Scheduling Example: LP Formulation

Let $(x_{\text{Mon}}, x_{\text{Tue}}, x_{\text{Wed}}, x_{\text{Thu}}, x_{\text{Fri}}, x_{\text{Sat}}, x_{\text{Sun}})$ denote the number of full-time employees whose 5-day workweek begins on the indicated day (each employee works 5 consecutive days, then has 2 consecutive days off). The goal is to minimize the total number of employees started this week subject to daily staffing requirements and nonnegativity:

Work Scheduling Example: LP Formulation

Let $(x_{\text{Mon}}, x_{\text{Tue}}, x_{\text{Wed}}, x_{\text{Thu}}, x_{\text{Fri}}, x_{\text{Sat}}, x_{\text{Sun}})$ denote the number of full-time employees whose 5-day workweek begins on the indicated day (each employee works 5 consecutive days, then has 2 consecutive days off). The goal is to minimize the total number of employees started this week subject to daily staffing requirements and nonnegativity:

$$\min \quad x_{\text{Mon}} + x_{\text{Tue}} + x_{\text{Wed}} + x_{\text{Thu}} + x_{\text{Fri}} + x_{\text{Sat}} + x_{\text{Sun}}$$

Work Scheduling Example: LP Formulation

Let $(x_{\text{Mon}}, x_{\text{Tue}}, x_{\text{Wed}}, x_{\text{Thu}}, x_{\text{Fri}}, x_{\text{Sat}}, x_{\text{Sun}})$ denote the number of full-time employees whose 5-day workweek begins on the indicated day (each employee works 5 consecutive days, then has 2 consecutive days off). The goal is to minimize the total number of employees started this week subject to daily staffing requirements and nonnegativity:

$$\begin{aligned} \min \quad & x_{\text{Mon}} + x_{\text{Tue}} + x_{\text{Wed}} + x_{\text{Thu}} + x_{\text{Fri}} + x_{\text{Sat}} + x_{\text{Sun}} \\ \text{s.t.} \quad & x_{\text{Mon}} + x_{\text{Sun}} + x_{\text{Sat}} + x_{\text{Fri}} + x_{\text{Thu}} \geq 7 \qquad \text{(Mon)} \end{aligned}$$

Work Scheduling Example: LP Formulation

Let $(x_{\text{Mon}}, x_{\text{Tue}}, x_{\text{Wed}}, x_{\text{Thu}}, x_{\text{Fri}}, x_{\text{Sat}}, x_{\text{Sun}})$ denote the number of full-time employees whose 5-day workweek begins on the indicated day (each employee works 5 consecutive days, then has 2 consecutive days off). The goal is to minimize the total number of employees started this week subject to daily staffing requirements and nonnegativity:

$$
\begin{aligned}
\min \quad & x_{\text{Mon}} + x_{\text{Tue}} + x_{\text{Wed}} + x_{\text{Thu}} + x_{\text{Fri}} + x_{\text{Sat}} + x_{\text{Sun}} & \\
\text{s.t.} \quad & x_{\text{Mon}} + x_{\text{Sun}} + x_{\text{Sat}} + x_{\text{Fri}} + x_{\text{Thu}} \geq 7 & \text{(Mon)} \\
& x_{\text{Tue}} + x_{\text{Mon}} + x_{\text{Sun}} + x_{\text{Sat}} + x_{\text{Fri}} \geq 3 & \text{(Tue)}
\end{aligned}
$$

Work Scheduling Example: LP Formulation

Let $(x_{\text{Mon}}, x_{\text{Tue}}, x_{\text{Wed}}, x_{\text{Thu}}, x_{\text{Fri}}, x_{\text{Sat}}, x_{\text{Sun}})$ denote the number of full-time employees whose 5-day workweek begins on the indicated day (each employee works 5 consecutive days, then has 2 consecutive days off). The goal is to minimize the total number of employees started this week subject to daily staffing requirements and nonnegativity:

$$
\begin{aligned}
\min \quad & x_{\text{Mon}} + x_{\text{Tue}} + x_{\text{Wed}} + x_{\text{Thu}} + x_{\text{Fri}} + x_{\text{Sat}} + x_{\text{Sun}} & \\
\text{s.t.} \quad & x_{\text{Mon}} + x_{\text{Sun}} + x_{\text{Sat}} + x_{\text{Fri}} + x_{\text{Thu}} \geq 7 & \text{(Mon)} \\
& x_{\text{Tue}} + x_{\text{Mon}} + x_{\text{Sun}} + x_{\text{Sat}} + x_{\text{Fri}} \geq 3 & \text{(Tue)} \\
& x_{\text{Wed}} + x_{\text{Tue}} + x_{\text{Mon}} + x_{\text{Sun}} + x_{\text{Sat}} \geq 5 & \text{(Wed)}
\end{aligned}
$$

Work Scheduling Example: LP Formulation

Let $(x_{\text{Mon}}, x_{\text{Tue}}, x_{\text{Wed}}, x_{\text{Thu}}, x_{\text{Fri}}, x_{\text{Sat}}, x_{\text{Sun}})$ denote the number of full-time employees whose 5-day workweek begins on the indicated day (each employee works 5 consecutive days, then has 2 consecutive days off). The goal is to minimize the total number of employees started this week subject to daily staffing requirements and nonnegativity:

$$
\begin{aligned}
\min \quad & x_{\text{Mon}} + x_{\text{Tue}} + x_{\text{Wed}} + x_{\text{Thu}} + x_{\text{Fri}} + x_{\text{Sat}} + x_{\text{Sun}} & \\
\text{s.t.} \quad & x_{\text{Mon}} + x_{\text{Sun}} + x_{\text{Sat}} + x_{\text{Fri}} + x_{\text{Thu}} \geq 7 & \text{(Mon)} \\
& x_{\text{Tue}} + x_{\text{Mon}} + x_{\text{Sun}} + x_{\text{Sat}} + x_{\text{Fri}} \geq 3 & \text{(Tue)} \\
& x_{\text{Wed}} + x_{\text{Tue}} + x_{\text{Mon}} + x_{\text{Sun}} + x_{\text{Sat}} \geq 5 & \text{(Wed)} \\
& x_{\text{Thu}} + x_{\text{Wed}} + x_{\text{Tue}} + x_{\text{Mon}} + x_{\text{Sun}} \geq 9 & \text{(Thu)}
\end{aligned}
$$

Work Scheduling Example: LP Formulation

Let $(x_{\text{Mon}}, x_{\text{Tue}}, x_{\text{Wed}}, x_{\text{Thu}}, x_{\text{Fri}}, x_{\text{Sat}}, x_{\text{Sun}})$ denote the number of full-time employees whose 5-day workweek begins on the indicated day (each employee works 5 consecutive days, then has 2 consecutive days off). The goal is to minimize the total number of employees started this week subject to daily staffing requirements and nonnegativity:

$$
\begin{aligned}
\min \quad & x_{\text{Mon}} + x_{\text{Tue}} + x_{\text{Wed}} + x_{\text{Thu}} + x_{\text{Fri}} + x_{\text{Sat}} + x_{\text{Sun}} & \\
\text{s.t.} \quad & x_{\text{Mon}} + x_{\text{Sun}} + x_{\text{Sat}} + x_{\text{Fri}} + x_{\text{Thu}} \geq 7 & \text{(Mon)} \\
& x_{\text{Tue}} + x_{\text{Mon}} + x_{\text{Sun}} + x_{\text{Sat}} + x_{\text{Fri}} \geq 3 & \text{(Tue)} \\
& x_{\text{Wed}} + x_{\text{Tue}} + x_{\text{Mon}} + x_{\text{Sun}} + x_{\text{Sat}} \geq 5 & \text{(Wed)} \\
& x_{\text{Thu}} + x_{\text{Wed}} + x_{\text{Tue}} + x_{\text{Mon}} + x_{\text{Sun}} \geq 9 & \text{(Thu)} \\
& x_{\text{Fri}} + x_{\text{Thu}} + x_{\text{Wed}} + x_{\text{Tue}} + x_{\text{Mon}} \geq 4 & \text{(Fri)}
\end{aligned}
$$

Work Scheduling Example: LP Formulation

Let $(x_{\text{Mon}}, x_{\text{Tue}}, x_{\text{Wed}}, x_{\text{Thu}}, x_{\text{Fri}}, x_{\text{Sat}}, x_{\text{Sun}})$ denote the number of full-time employees whose 5-day workweek begins on the indicated day (each employee works 5 consecutive days, then has 2 consecutive days off). The goal is to minimize the total number of employees started this week subject to daily staffing requirements and nonnegativity:

$$
\begin{aligned}
\min \quad & x_{\text{Mon}} + x_{\text{Tue}} + x_{\text{Wed}} + x_{\text{Thu}} + x_{\text{Fri}} + x_{\text{Sat}} + x_{\text{Sun}} & \\
\text{s.t.} \quad & x_{\text{Mon}} + x_{\text{Sun}} + x_{\text{Sat}} + x_{\text{Fri}} + x_{\text{Thu}} \geq 7 & \text{(Mon)} \\
& x_{\text{Tue}} + x_{\text{Mon}} + x_{\text{Sun}} + x_{\text{Sat}} + x_{\text{Fri}} \geq 3 & \text{(Tue)} \\
& x_{\text{Wed}} + x_{\text{Tue}} + x_{\text{Mon}} + x_{\text{Sun}} + x_{\text{Sat}} \geq 5 & \text{(Wed)} \\
& x_{\text{Thu}} + x_{\text{Wed}} + x_{\text{Tue}} + x_{\text{Mon}} + x_{\text{Sun}} \geq 9 & \text{(Thu)} \\
& x_{\text{Fri}} + x_{\text{Thu}} + x_{\text{Wed}} + x_{\text{Tue}} + x_{\text{Mon}} \geq 4 & \text{(Fri)} \\
& x_{\text{Sat}} + x_{\text{Fri}} + x_{\text{Thu}} + x_{\text{Wed}} + x_{\text{Tue}} \geq 6 & \text{(Sat)}
\end{aligned}
$$

Work Scheduling Example: LP Formulation

Let $(x_{\text{Mon}}, x_{\text{Tue}}, x_{\text{Wed}}, x_{\text{Thu}}, x_{\text{Fri}}, x_{\text{Sat}}, x_{\text{Sun}})$ denote the number of full-time employees whose 5-day workweek begins on the indicated day (each employee works 5 consecutive days, then has 2 consecutive days off). The goal is to minimize the total number of employees started this week subject to daily staffing requirements and nonnegativity:

$$
\begin{aligned}
\min \quad & x_{\text{Mon}} + x_{\text{Tue}} + x_{\text{Wed}} + x_{\text{Thu}} + x_{\text{Fri}} + x_{\text{Sat}} + x_{\text{Sun}} & \\
\text{s.t.} \quad & x_{\text{Mon}} + x_{\text{Sun}} + x_{\text{Sat}} + x_{\text{Fri}} + x_{\text{Thu}} \geq 7 & \text{(Mon)} \\
& x_{\text{Tue}} + x_{\text{Mon}} + x_{\text{Sun}} + x_{\text{Sat}} + x_{\text{Fri}} \geq 3 & \text{(Tue)} \\
& x_{\text{Wed}} + x_{\text{Tue}} + x_{\text{Mon}} + x_{\text{Sun}} + x_{\text{Sat}} \geq 5 & \text{(Wed)} \\
& x_{\text{Thu}} + x_{\text{Wed}} + x_{\text{Tue}} + x_{\text{Mon}} + x_{\text{Sun}} \geq 9 & \text{(Thu)} \\
& x_{\text{Fri}} + x_{\text{Thu}} + x_{\text{Wed}} + x_{\text{Tue}} + x_{\text{Mon}} \geq 4 & \text{(Fri)} \\
& x_{\text{Sat}} + x_{\text{Fri}} + x_{\text{Thu}} + x_{\text{Wed}} + x_{\text{Tue}} \geq 6 & \text{(Sat)} \\
& x_{\text{Sun}} + x_{\text{Sat}} + x_{\text{Fri}} + x_{\text{Thu}} + x_{\text{Wed}} \geq 1 & \text{(Sun)}
\end{aligned}
$$

Work Scheduling Example: LP Formulation

Let $(x_{\text{Mon}}, x_{\text{Tue}}, x_{\text{Wed}}, x_{\text{Thu}}, x_{\text{Fri}}, x_{\text{Sat}}, x_{\text{Sun}})$ denote the number of full-time employees whose 5-day workweek begins on the indicated day (each employee works 5 consecutive days, then has 2 consecutive days off). The goal is to minimize the total number of employees started this week subject to daily staffing requirements and nonnegativity:

$$
\begin{aligned}
\min \quad & x_{\text{Mon}} + x_{\text{Tue}} + x_{\text{Wed}} + x_{\text{Thu}} + x_{\text{Fri}} + x_{\text{Sat}} + x_{\text{Sun}} & \\
\text{s.t.} \quad & x_{\text{Mon}} + x_{\text{Sun}} + x_{\text{Sat}} + x_{\text{Fri}} + x_{\text{Thu}} \geq 7 & \text{(Mon)} \\
& x_{\text{Tue}} + x_{\text{Mon}} + x_{\text{Sun}} + x_{\text{Sat}} + x_{\text{Fri}} \geq 3 & \text{(Tue)} \\
& x_{\text{Wed}} + x_{\text{Tue}} + x_{\text{Mon}} + x_{\text{Sun}} + x_{\text{Sat}} \geq 5 & \text{(Wed)} \\
& x_{\text{Thu}} + x_{\text{Wed}} + x_{\text{Tue}} + x_{\text{Mon}} + x_{\text{Sun}} \geq 9 & \text{(Thu)} \\
& x_{\text{Fri}} + x_{\text{Thu}} + x_{\text{Wed}} + x_{\text{Tue}} + x_{\text{Mon}} \geq 4 & \text{(Fri)} \\
& x_{\text{Sat}} + x_{\text{Fri}} + x_{\text{Thu}} + x_{\text{Wed}} + x_{\text{Tue}} \geq 6 & \text{(Sat)} \\
& x_{\text{Sun}} + x_{\text{Sat}} + x_{\text{Fri}} + x_{\text{Thu}} + x_{\text{Wed}} \geq 1 & \text{(Sun)} \\
& x_{\text{Mon}}, x_{\text{Tue}}, x_{\text{Wed}}, x_{\text{Thu}}, x_{\text{Fri}}, x_{\text{Sat}}, x_{\text{Sun}} \geq 0. &
\end{aligned}
$$

Work Scheduling Solution (PuLP) (1/2)

```
import pulp as pl

# Create the problem
prob = pl.LpProblem("Post_Office_Scheduling", pl.LpMinimize)

# Define variables directly into the model
x_mon = prob.add_variable("x_Mon", lowBound=0, cat='Integer')
x_tue = prob.add_variable("x_Tue", lowBound=0, cat='Integer')
x_wed = prob.add_variable("x_Wed", lowBound=0, cat='Integer')
x_thu = prob.add_variable("x_Thu", lowBound=0, cat='Integer')
x_fri = prob.add_variable("x_Fri", lowBound=0, cat='Integer')
x_sat = prob.add_variable("x_Sat", lowBound=0, cat='Integer')
x_sun = prob.add_variable("x_Sun", lowBound=0, cat='Integer')

# Objective function
prob += x_mon + x_tue + x_wed + x_thu + x_fri + x_sat + x_sun, "Total_Employees"

# Constraints
prob += x_mon + x_sun + x_sat + x_fri + x_thu >= 7, "Monday"
prob += x_tue + x_mon + x_sun + x_sat + x_fri >= 3, "Tuesday"
prob += x_wed + x_tue + x_mon + x_sun + x_sat >= 5, "Wednesday"
prob += x_thu + x_wed + x_tue + x_mon + x_sun >= 9, "Thursday"
prob += x_fri + x_thu + x_wed + x_tue + x_mon >= 4, "Friday"
prob += x_sat + x_fri + x_thu + x_wed + x_tue >= 6, "Saturday"
prob += x_sun + x_sat + x_fri + x_thu + x_wed >= 1, "Sunday"
```

Work Scheduling Solution (PuLP) (2/2)

```
# Solve the problem
res = prob.solve()

# Print results
print("Status:", res.status.name)
print("Optimal Solution:")
print(f"Employees starting Mon = {x_mon.varValue}")
print(f"Employees starting Tue = {x_tue.varValue}")
print(f"Employees starting Wed = {x_wed.varValue}")
print(f"Employees starting Thu = {x_thu.varValue}")
print(f"Employees starting Fri = {x_fri.varValue}")
print(f"Employees starting Sat = {x_sat.varValue}")
print(f"Employees starting Sun = {x_sun.varValue}")
print(f"Minimum Employees = {pl.value(prob.objective)}")
```

Blending Example II

The Java Junkie Coffee Shop blends coffee on the premises for its customers. It sells three basic blends in 1-pound bags: Special, Mountain Dark, and Mill Regular. It uses two different types of beans to produce the blends: Brazilian and mild. The shop used the following blend recipe requirements.

BLEND	MIX REQUIREMENTS	SELLING PRICE ($/LB)
Special	At least 40% Brazilian No more 30% mild	6.50
Dark	At least 60% Brazilian No more than 10% mild	5.25
Regular	No more than 60% mild At least 30% Brazilian	3.75

The cost of Brazilian coffee beans are $2.00 per pound and the cost of mild is $1.25 per pound. Every week the shop can buy up-to 210 pounds of Brazilian coffee beans, and 300 pounds of mild coffee beans. The shop wants to know the amount of each blend it should prepare each week and how much of each type of coffee beans to include in each blend in order to maximize profits. Assume the weekly demand to all three blends always exceeds the store's supply.

Blending Example II: LP with Double-Indexed Variables

Let (x_{bS}, x_{bD}, x_{bR}) denote pounds of Brazilian beans used in Special, Dark, and Regular, and (x_{mS}, x_{mD}, x_{mR}) denote pounds of mild beans used in those blends. The objective maximizes profit (revenue minus bean costs), with recipe, availability, and nonnegativity constraints:

$$\max \quad 6.50(x_{bS} + x_{mS}) + 5.25(x_{bD} + x_{mD}) + 3.75(x_{bR} + x_{mR})$$

Blending Example II: LP with Double-Indexed Variables

Let (x_{bS}, x_{bD}, x_{bR}) denote pounds of Brazilian beans used in Special, Dark, and Regular, and (x_{mS}, x_{mD}, x_{mR}) denote pounds of mild beans used in those blends. The objective maximizes profit (revenue minus bean costs), with recipe, availability, and nonnegativity constraints:

$$\begin{aligned}\max \quad & 6.50(x_{bS} + x_{mS}) + 5.25(x_{bD} + x_{mD}) + 3.75(x_{bR} + x_{mR}) \\ & - 2.00(x_{bS} + x_{bD} + x_{bR}) - 1.25(x_{mS} + x_{mD} + x_{mR})\end{aligned}$$

Blending Example II: LP with Double-Indexed Variables

Let (x_{bS}, x_{bD}, x_{bR}) denote pounds of Brazilian beans used in Special, Dark, and Regular, and (x_{mS}, x_{mD}, x_{mR}) denote pounds of mild beans used in those blends. The objective maximizes profit (revenue minus bean costs), with recipe, availability, and nonnegativity constraints:

$$\begin{aligned}
\max \quad & 6.50(x_{bS} + x_{mS}) + 5.25(x_{bD} + x_{mD}) + 3.75(x_{bR} + x_{mR}) \\
& - 2.00(x_{bS} + x_{bD} + x_{bR}) - 1.25(x_{mS} + x_{mD} + x_{mR}) \\
\text{s.t.} \quad & \text{Special: } x_{bS} \geq 0.40(x_{bS} + x_{mS}), \quad x_{mS} \leq 0.30(x_{bS} + x_{mS}),
\end{aligned}$$

Blending Example II: LP with Double-Indexed Variables

Let (x_{bS}, x_{bD}, x_{bR}) denote pounds of Brazilian beans used in Special, Dark, and Regular, and (x_{mS}, x_{mD}, x_{mR}) denote pounds of mild beans used in those blends. The objective maximizes profit (revenue minus bean costs), with recipe, availability, and nonnegativity constraints:

$$\begin{aligned}
\max \quad & 6.50(x_{bS} + x_{mS}) + 5.25(x_{bD} + x_{mD}) + 3.75(x_{bR} + x_{mR}) \\
& - 2.00(x_{bS} + x_{bD} + x_{bR}) - 1.25(x_{mS} + x_{mD} + x_{mR}) \\
\text{s.t.} \quad & \text{Special: } x_{bS} \geq 0.40(x_{bS} + x_{mS}), \quad x_{mS} \leq 0.30(x_{bS} + x_{mS}), \\
& \text{Dark: } x_{bD} \geq 0.60(x_{bD} + x_{mD}), \quad x_{mD} \leq 0.10(x_{bD} + x_{mD}),
\end{aligned}$$

Blending Example II: LP with Double-Indexed Variables

Let (x_{bS}, x_{bD}, x_{bR}) denote pounds of Brazilian beans used in Special, Dark, and Regular, and (x_{mS}, x_{mD}, x_{mR}) denote pounds of mild beans used in those blends. The objective maximizes profit (revenue minus bean costs), with recipe, availability, and nonnegativity constraints:

$$\begin{aligned}
\max \quad & 6.50(x_{bS} + x_{mS}) + 5.25(x_{bD} + x_{mD}) + 3.75(x_{bR} + x_{mR}) \\
& - 2.00(x_{bS} + x_{bD} + x_{bR}) - 1.25(x_{mS} + x_{mD} + x_{mR}) \\
\text{s.t.} \quad & \text{Special: } x_{bS} \geq 0.40(x_{bS} + x_{mS}), \quad x_{mS} \leq 0.30(x_{bS} + x_{mS}), \\
& \text{Dark: } x_{bD} \geq 0.60(x_{bD} + x_{mD}), \quad x_{mD} \leq 0.10(x_{bD} + x_{mD}), \\
& \text{Regular: } x_{mR} \leq 0.60(x_{bR} + x_{mR}), \quad x_{bR} \geq 0.30(x_{bR} + x_{mR}),
\end{aligned}$$

Blending Example II: LP with Double-Indexed Variables

Let (x_{bS}, x_{bD}, x_{bR}) denote pounds of Brazilian beans used in Special, Dark, and Regular, and (x_{mS}, x_{mD}, x_{mR}) denote pounds of mild beans used in those blends. The objective maximizes profit (revenue minus bean costs), with recipe, availability, and nonnegativity constraints:

$$\begin{aligned}
\max \quad & 6.50(x_{bS} + x_{mS}) + 5.25(x_{bD} + x_{mD}) + 3.75(x_{bR} + x_{mR}) \\
& - 2.00(x_{bS} + x_{bD} + x_{bR}) - 1.25(x_{mS} + x_{mD} + x_{mR}) \\
\text{s.t.} \quad & \text{Special: } x_{bS} \geq 0.40(x_{bS} + x_{mS}), \quad x_{mS} \leq 0.30(x_{bS} + x_{mS}), \\
& \text{Dark: } x_{bD} \geq 0.60(x_{bD} + x_{mD}), \quad x_{mD} \leq 0.10(x_{bD} + x_{mD}), \\
& \text{Regular: } x_{mR} \leq 0.60(x_{bR} + x_{mR}), \quad x_{bR} \geq 0.30(x_{bR} + x_{mR}), \\
& \text{Brazilian availability: } x_{bS} + x_{bD} + x_{bR} \leq 210,
\end{aligned}$$

Blending Example II: LP with Double-Indexed Variables

Let (x_{bS}, x_{bD}, x_{bR}) denote pounds of Brazilian beans used in Special, Dark, and Regular, and (x_{mS}, x_{mD}, x_{mR}) denote pounds of mild beans used in those blends. The objective maximizes profit (revenue minus bean costs), with recipe, availability, and nonnegativity constraints:

$$
\begin{aligned}
\max \quad & 6.50(x_{bS} + x_{mS}) + 5.25(x_{bD} + x_{mD}) + 3.75(x_{bR} + x_{mR}) \\
& \quad - 2.00(x_{bS} + x_{bD} + x_{bR}) - 1.25(x_{mS} + x_{mD} + x_{mR}) \\
\text{s.t.} \quad & \text{Special: } x_{bS} \geq 0.40(x_{bS} + x_{mS}), \quad x_{mS} \leq 0.30(x_{bS} + x_{mS}), \\
& \text{Dark: } x_{bD} \geq 0.60(x_{bD} + x_{mD}), \quad x_{mD} \leq 0.10(x_{bD} + x_{mD}), \\
& \text{Regular: } x_{mR} \leq 0.60(x_{bR} + x_{mR}), \quad x_{bR} \geq 0.30(x_{bR} + x_{mR}), \\
& \text{Brazilian availability: } x_{bS} + x_{bD} + x_{bR} \leq 210, \\
& \text{Mild availability: } x_{mS} + x_{mD} + x_{mR} \leq 300,
\end{aligned}
$$

Blending Example II: LP with Double-Indexed Variables

Let (x_{bS}, x_{bD}, x_{bR}) denote pounds of Brazilian beans used in Special, Dark, and Regular, and (x_{mS}, x_{mD}, x_{mR}) denote pounds of mild beans used in those blends. The objective maximizes profit (revenue minus bean costs), with recipe, availability, and nonnegativity constraints:

$$
\begin{aligned}
\max \quad & 6.50(x_{bS} + x_{mS}) + 5.25(x_{bD} + x_{mD}) + 3.75(x_{bR} + x_{mR}) \\
& - 2.00(x_{bS} + x_{bD} + x_{bR}) - 1.25(x_{mS} + x_{mD} + x_{mR}) \\
\text{s.t.} \quad & \text{Special: } x_{bS} \geq 0.40(x_{bS} + x_{mS}), \quad x_{mS} \leq 0.30(x_{bS} + x_{mS}), \\
& \text{Dark: } x_{bD} \geq 0.60(x_{bD} + x_{mD}), \quad x_{mD} \leq 0.10(x_{bD} + x_{mD}), \\
& \text{Regular: } x_{mR} \leq 0.60(x_{bR} + x_{mR}), \quad x_{bR} \geq 0.30(x_{bR} + x_{mR}), \\
& \text{Brazilian availability: } x_{bS} + x_{bD} + x_{bR} \leq 210, \\
& \text{Mild availability: } x_{mS} + x_{mD} + x_{mR} \leq 300, \\
& x_{bS}, x_{bD}, x_{bR}, x_{mS}, x_{mD}, x_{mR} \geq 0.
\end{aligned}
$$

Equivalently, the recipe constraints may be written in purely linear form:

$$
\begin{aligned}
& 0.60x_{bS} - 0.40x_{mS} \geq 0, \quad -0.30x_{bS} + 0.70x_{mS} \leq 0, \\
& 0.40x_{bD} - 0.60x_{mD} \geq 0, \quad -0.10x_{bD} + 0.90x_{mD} \leq 0, \\
& -0.60x_{bR} + 0.40x_{mR} \leq 0, \quad 0.70x_{bR} - 0.30x_{mR} \geq 0.
\end{aligned}
$$

Coffee Blending Solution (PuLP) (1/2)

```
import pulp as pl

# Create the problem
prob = pl.LpProblem("Java_Junkie_Coffee", pl.LpMaximize)

# Define variables directly into the model
x_bS = prob.add_variable("Brazilian_Special", lowBound=0)
x_bD = prob.add_variable("Brazilian_Dark", lowBound=0)
x_bR = prob.add_variable("Brazilian_Regular", lowBound=0)
x_mS = prob.add_variable("Mild_Special", lowBound=0)
x_mD = prob.add_variable("Mild_Dark", lowBound=0)
x_mR = prob.add_variable("Mild_Regular", lowBound=0)

# Objective function
prob += (6.50 * (x_bS + x_mS) + 5.25 * (x_bD + x_mD) + 3.75 * (x_bR + x_mR)
         - 2.00 * (x_bS + x_bD + x_bR) - 1.25 * (x_mS + x_mD + x_mR)), "Maximize_Profit"

# Constraints
# Special blend constraints
prob += x_bS >= 0.40 * (x_bS + x_mS), "Special_Brazilian_Min"
prob += x_mS <= 0.30 * (x_bS + x_mS), "Special_Mild_Max"

# Dark blend constraints
prob += x_bD >= 0.60 * (x_bD + x_mD), "Dark_Brazilian_Min"
prob += x_mD <= 0.10 * (x_bD + x_mD), "Dark_Mild_Max"

# Regular blend constraints
prob += x_mR <= 0.60 * (x_bR + x_mR), "Regular_Mild_Max"
prob += x_bR >= 0.30 * (x_bR + x_mR), "Regular_Brazilian_Min"

# Availability constraints
prob += x_bS + x_bD + x_bR <= 210, "Brazilian_Availability"
prob += x_mS + x_mD + x_mR <= 300, "Mild_Availability"
```

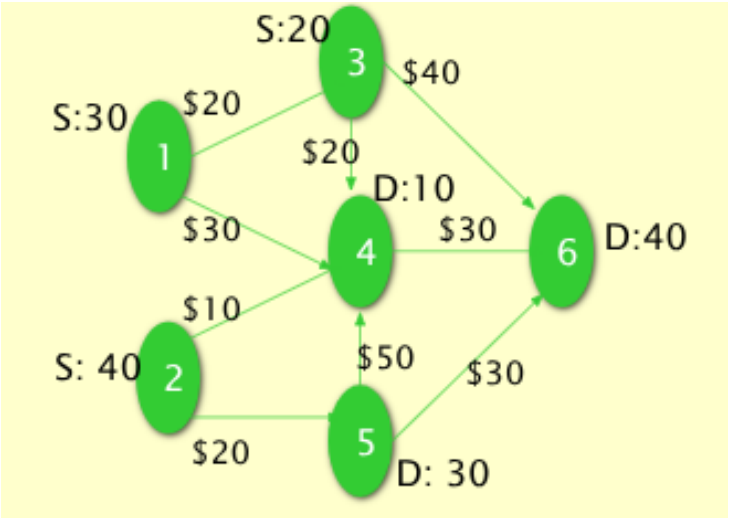
Coffee Blending Solution (PuLP) (2/2)

```
# Solve the problem
res = prob.solve()

# Print results
print("Status:", res.status.name)
print("Optimal Solution:")
print(f"Brazilian in Special = {x_bS.varValue} lbs")
print(f"Brazilian in Dark = {x_bD.varValue} lbs")
print(f"Brazilian in Regular = {x_bR.varValue} lbs")
print(f"Mild in Special = {x_mS.varValue} lbs")
print(f"Mild in Dark = {x_mD.varValue} lbs")
print(f"Mild in Regular = {x_mR.varValue} lbs")
print(f"Maximum Profit = ${pl.value(prob.objective)}")
```

Network Model: Transportation Problem

A furniture manufacturer, Furniture Max, has three warehouses (1, 2, 3) , and three retail stores (4, 5, 6). The following network diagram indicates the shipping routes and shipping costs. The values on the arcs indicate the per unit shipping costs required to transport furniture between the various cities. The supply of furniture at each warehouse is indicated by the number next to nodes 1, 2, and 3. The demand for furniture is indicated by the number next to the nodes 4, 5, 6. Formulate an LP to determine the least costly shipping plan for Furniture Max.



Transportation Problem

Decisions x_{ij} = units shipped from node i to node j

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Objective

$$\min \ 20x_{13} + 30x_{14} + \cdots + 30x_{56}$$

Transportation Problem

Decisions x_{ij} = units shipped from node i to node j

Objective

$$\min \ 20x_{13} + 30x_{14} + \cdots + 30x_{56}$$

Node 1: $x_{13} + x_{14} \leq 30$

Transportation Problem

Decisions x_{ij} = units shipped from node i to node j

Objective

$$\min \ 20x_{13} + 30x_{14} + \cdots + 30x_{56}$$

Node 1: $x_{13} + x_{14} \leq 30$

Node 2: $x_{24} + x_{25} \leq 40$

Transportation Problem

Decisions x_{ij} = units shipped from node i to node j

Objective

$$\min \ 20x_{13} + 30x_{14} + \cdots + 30x_{56}$$

$$\text{Node 1: } x_{13} + x_{14} \leq 30$$

$$\text{Node 2: } x_{24} + x_{25} \leq 40$$

$$\text{Node 3: } x_{36} + x_{34} - x_{13} \leq 20$$

Transportation Problem

Decisions x_{ij} = units shipped from node i to node j

Objective

$$\min \; 20x_{13} + 30x_{14} + \cdots + 30x_{56}$$

$$\text{Node 1: } x_{13} + x_{14} \leq 30$$

$$\text{Node 2: } x_{24} + x_{25} \leq 40$$

$$\text{Node 3: } x_{36} + x_{34} - x_{13} \leq 20$$

$$\text{Node 4: } x_{14} + x_{24} + x_{34} + x_{54} - x_{46} \geq 10$$

Transportation Problem

Decisions x_{ij} = units shipped from node i to node j

Objective

$$\min \; 20x_{13} + 30x_{14} + \cdots + 30x_{56}$$

$$\text{Node 1: } x_{13} + x_{14} \leq 30$$

$$\text{Node 2: } x_{24} + x_{25} \leq 40$$

$$\text{Node 3: } x_{36} + x_{34} - x_{13} \leq 20$$

$$\text{Node 4: } x_{14} + x_{24} + x_{34} + x_{54} - x_{46} \geq 10$$

$$\text{Node 5: } x_{25} - x_{54} - x_{56} \geq 30$$

Transportation Problem

Decisions x_{ij} = units shipped from node i to node j

Objective

$$\min \; 20x_{13} + 30x_{14} + \cdots + 30x_{56}$$

$$\text{Node 1: } x_{13} + x_{14} \leq 30$$

$$\text{Node 2: } x_{24} + x_{25} \leq 40$$

$$\text{Node 3: } x_{36} + x_{34} - x_{13} \leq 20$$

$$\text{Node 4: } x_{14} + x_{24} + x_{34} + x_{54} - x_{46} \geq 10$$

$$\text{Node 5: } x_{25} - x_{54} - x_{56} \geq 30$$

$$\text{Node 6: } x_{36} + x_{56} + x_{46} \geq 40$$

Transportation Problem

Decisions x_{ij} = units shipped from node i to node j

Objective

$$\min \; 20x_{13} + 30x_{14} + \cdots + 30x_{56}$$

$$\text{Node 1: } x_{13} + x_{14} \leq 30$$

$$\text{Node 2: } x_{24} + x_{25} \leq 40$$

$$\text{Node 3: } x_{36} + x_{34} - x_{13} \leq 20$$

$$\text{Node 4: } x_{14} + x_{24} + x_{34} + x_{54} - x_{46} \geq 10$$

$$\text{Node 5: } x_{25} - x_{54} - x_{56} \geq 30$$

$$\text{Node 6: } x_{36} + x_{56} + x_{46} \geq 40$$

$$x_{ij} \geq 0$$

Transportation Problem Solution (PuLP) (1/2)

```
import pulp as pl

# Create the problem
prob = pl.LpProblem("Furniture_Max_Transportation", pl.LpMinimize)

# Define variables directly into the model
x13 = prob.add_variable("x13", lowBound=0)
x14 = prob.add_variable("x14", lowBound=0)
x24 = prob.add_variable("x24", lowBound=0)
x25 = prob.add_variable("x25", lowBound=0)
x34 = prob.add_variable("x34", lowBound=0)
x36 = prob.add_variable("x36", lowBound=0)
x46 = prob.add_variable("x46", lowBound=0)
x54 = prob.add_variable("x54", lowBound=0)
x56 = prob.add_variable("x56", lowBound=0)

# Objective function
prob += 20 * x13 + 30 * x14 + 10 * x24 + 20 * x25 + 30 * x34 + 40 * x36 + 30 * x46 + 50 *
      x54 + 30 * x56, "Minimize_Cost"

# Constraints
# Supply constraints
prob += x13 + x14 <= 30, "Node_1_Supply"
prob += x24 + x25 <= 40, "Node_2_Supply"
prob += x34 + x36 - x13 <= 20, "Node_3_Supply"

# Demand constraints
prob += x14 + x24 + x34 + x54 - x46 >= 10, "Node_4_Demand"
prob += x25 - x54 - x56 >= 30, "Node_5_Demand"
prob += x36 + x56 + x46 >= 40, "Node_6_Demand"
```

Transportation Problem Solution (PuLP) (2/2)

```
# Solve the problem
res = prob.solve()

# Print results
print("Status:", res.status.name)
print("Optimal Solution:")
print(f"x13 = {x13.varValue}")
print(f"x14 = {x14.varValue}")
print(f"x24 = {x24.varValue}")
print(f"x25 = {x25.varValue}")
print(f"x34 = {x34.varValue}")
print(f"x36 = {x36.varValue}")
print(f"x46 = {x46.varValue}")
print(f"x54 = {x54.varValue}")
print(f"x56 = {x56.varValue}")
print(f"Minimum Cost = ${pl.value(prob.objective)}")
```

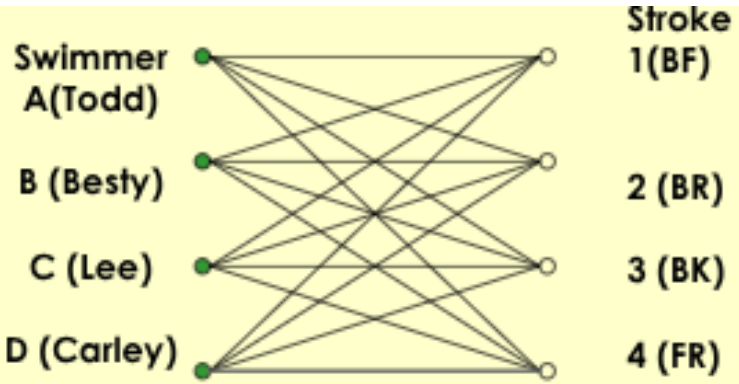
Network Model: Assignment Problem

Coach Kemppel is the coach of the Buchanan Swim Club's co-ed team. Her team competes against other swim clubs, and a perennial question for the coach is how to organize the medley relay team. The medley relay requires each of four swimmers to swim a different stroke: butterfly, breaststroke, backstroke, and freestyle. The relay is the final event in the competition, and the outcome of the swim meet often depends on the performance of the relay team.
During practice, Coach Kemppel has asked each of her top four swimmers to try each of the four stokes, and she has tracked their times (in seconds), as shown in the following table.

	Stroke			
	BF	BR	BK	FR
Todd	38	75	44	27
Besty	34	76	43	25
Lee	41	71	41	26
Carly	33	80	45	30

With this information, Coach Kemppel is ready to assign swimmers to stokes in the relay race, but she can see that a lot of combinations are possible. Formulate a linear programming model to help Coach Kemppel make this decision.

An Assignment Problem



- Decision Variables:

  x_{ij} =1 assign swimmer i to stroke j

  =0 not assign swimmer i to stroke j

  i=A, B, C,D, j=1,2,3,4

Medley Relay Assignment: LP Formulation

Let $I = \{\text{Todd}, \text{Besty}, \text{Lee}, \text{Carly}\}$ and $S = \{\text{BF}, \text{BR}, \text{BK}, \text{FR}\}$. For each swimmer $i \in I$ and stroke $s \in S$, let $x_{i,s}$ denote the assignment decision, with $x_{i,s} = 1$ indicating that swimmer i swims stroke s, and $x_{i,s} = 0$ otherwise. Let $c_{i,s}$ be the observed practice time (seconds) from the table. The model minimizes total relay time subject to one-to-one assignment:

$$\min_{x} \quad \sum_{i \in I} \sum_{s \in S} c_{i,s} x_{i,s}$$

$$\text{s.t.} \quad \sum_{s \in S} x_{i,s} = 1 \qquad \forall i \in I \quad \text{(each swimmer takes exactly one stroke)}$$

$$\sum_{i \in I} x_{i,s} = 1 \qquad \forall s \in S \quad \text{(each stroke assigned to exactly one swimmer)}$$

$$x_{i,s} \in \{0, 1\} \qquad \forall i \in I,\ \forall s \in S.$$

The cost coefficients $c_{i,s}$ are given by the measured times (seconds):
$c_{\text{Todd,BF}} = 38$, $c_{\text{Todd,BR}} = 75$, $c_{\text{Todd,BK}} = 44$, $c_{\text{Todd,FR}} = 27$;
$c_{\text{Besty,BF}} = 34$, $c_{\text{Besty,BR}} = 76$, $c_{\text{Besty,BK}} = 43$, $c_{\text{Besty,FR}} = 25$; $c_{\text{Lee,BF}} = 41$, $c_{\text{Lee,BR}} = 71$, $c_{\text{Lee,BK}} = 41$, $c_{\text{Lee,FR}} = 26$; $c_{\text{Carly,BF}} = 33$, $c_{\text{Carly,BR}} = 80$, $c_{\text{Carly,BK}} = 45$, $c_{\text{Carly,FR}} = 30$.

Medley Relay Assignment Solution (PuLP) (1/2)

```
import pulp as pl

# Create the problem
prob = pl.LpProblem("Medley_Relay_Assignment", pl.LpMinimize)

# Define swimmers and strokes
swimmers = ['Todd', 'Besty', 'Lee', 'Carly']
strokes = ['BF', 'BR', 'BK', 'FR']

# Cost coefficients (times)
costs = {
    'Todd': {'BF': 38, 'BR': 75, 'BK': 44, 'FR': 27},
    'Besty': {'BF': 34, 'BR': 76, 'BK': 43, 'FR': 25},
    'Lee': {'BF': 41, 'BR': 71, 'BK': 41, 'FR': 26},
    'Carly': {'BF': 33, 'BR': 80, 'BK': 45, 'FR': 30}
}

# Decision variables
x = prob.add_variable_dicts("Assign", (swimmers, strokes), lowBound=0, upBound=1, cat='
    Integer')

# Objective function
prob += pl.lpSum(costs[i][s] * x[i][s] for i in swimmers for s in strokes), "Total_Time"

# Constraints: Each swimmer assigned to exactly one stroke
for i in swimmers:
    prob += pl.lpSum(x[i][s] for s in strokes) == 1, f"Swimmer_{i}"

# Each stroke assigned to exactly one swimmer
for s in strokes:
    prob += pl.lpSum(x[i][s] for i in swimmers) == 1, f"Stroke_{s}"
```

Medley Relay Assignment Solution (PuLP) (2/2)

```
# Solve the problem
res = prob.solve()

# Print results
print("Status:", res.status.name)
print("Optimal Assignments:")
for i in swimmers:
    for s in strokes:
        if x[i][s].varValue == 1:
            print(f"{i} assigned to {s} (Time: {costs[i][s]})")
print(f"Minimum Total Time = {pl.value(prob.objective)} seconds")
```

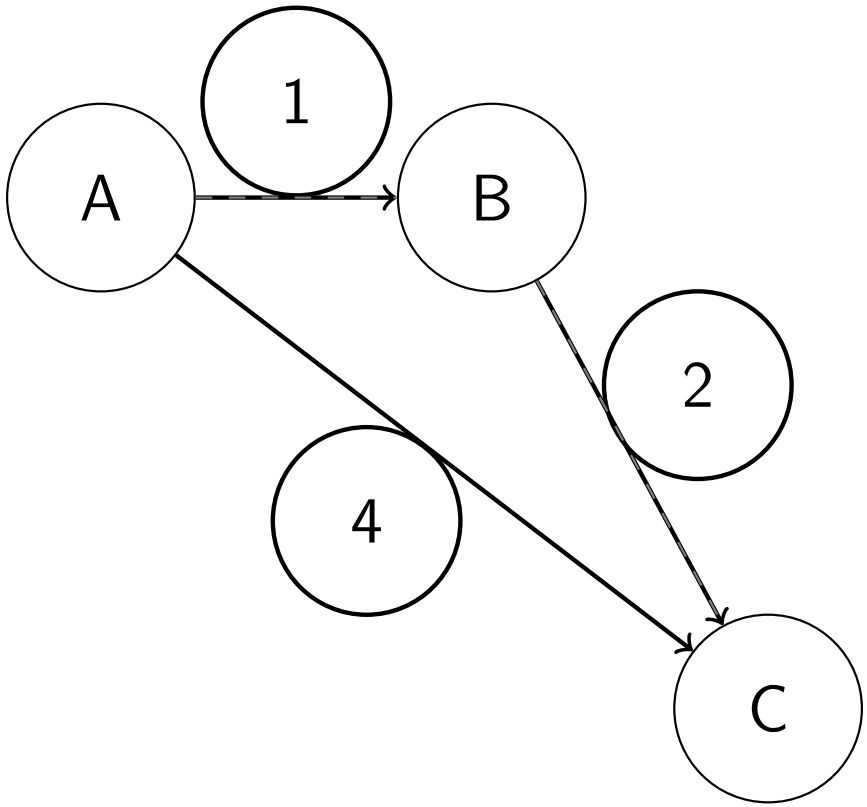
The Shortest Path Problem

LP and an Heuristic

- Using a toy example

The Shortest Path Problem: An Example

Find the shortest path from A to C.



Edges with costs:

- A to B (1)
- A to C (4)
- B to C (2)

Problem Statement

- The shortest path problem seeks the path from source (A) to sink (C) with minimal total cost.
- Traditional algorithms: Dijkstra, Bellman-Ford.
- Here, we formulate it as a Linear Program (LP) using network flow.

LP Formulation: Variables and Objective

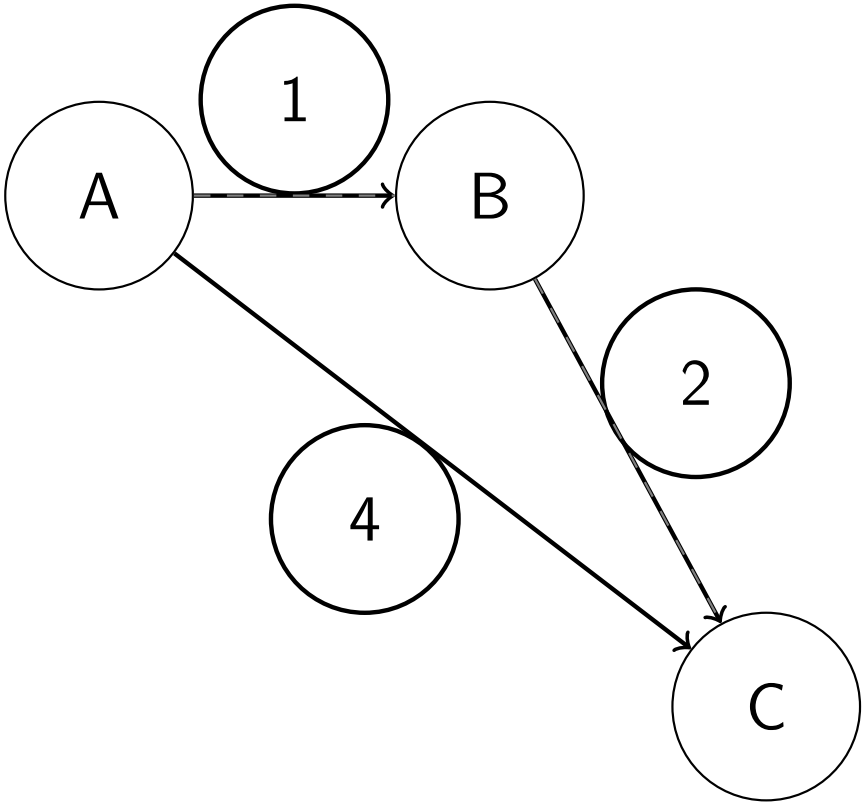
Variables (flow on edges):

- x_{AB}: flow from A to B
- x_{AC}: flow from A to C
- x_{BC}: flow from B to C

Objective: Minimize total cost

$$\min \ 1 \cdot x_{AB} + 4 \cdot x_{AC} + 2 \cdot x_{BC}$$

LP Formulation: Constraints



Flow conservation (net flow out):

- Source A: net out = 1 $\quad x_{AB} + x_{AC} = 1$
- Intermediate B: net out = 0 $\quad x_{BC} - x_{AB} = 0$
- Sink C: net out = -1 $\quad -(x_{AC} + x_{BC}) = -1$ or $x_{AC} + x_{BC} = 1$

Non-negativity: $x_{AB}, x_{AC}, x_{BC} \geq 0$

Solving with PuLP

```
import pulp as pl

prob = pl.LpProblem("ShortestPath", pl.LpMinimize)

x_ab = prob.add_variable("x_ab", lowBound=0)
x_ac = prob.add_variable("x_ac", lowBound=0)
x_bc = prob.add_variable("x_bc", lowBound=0)

prob += 1*x_ab + 4*x_ac + 2*x_bc

prob += x_ab + x_ac == 1, "Node_A"
prob += x_bc - x_ab == 0, "Node_B"
prob += x_ac + x_bc == 1, "Node_C"

res = prob.solve()
```

Optimal Solution

- $x_{AB} = 1$
- $x_{AC} = 0$
- $x_{BC} = 1$
- Total Cost $= 3$

Path: A $\rightarrow$ B $\rightarrow$ C (cost $1 + 2 = 3$)

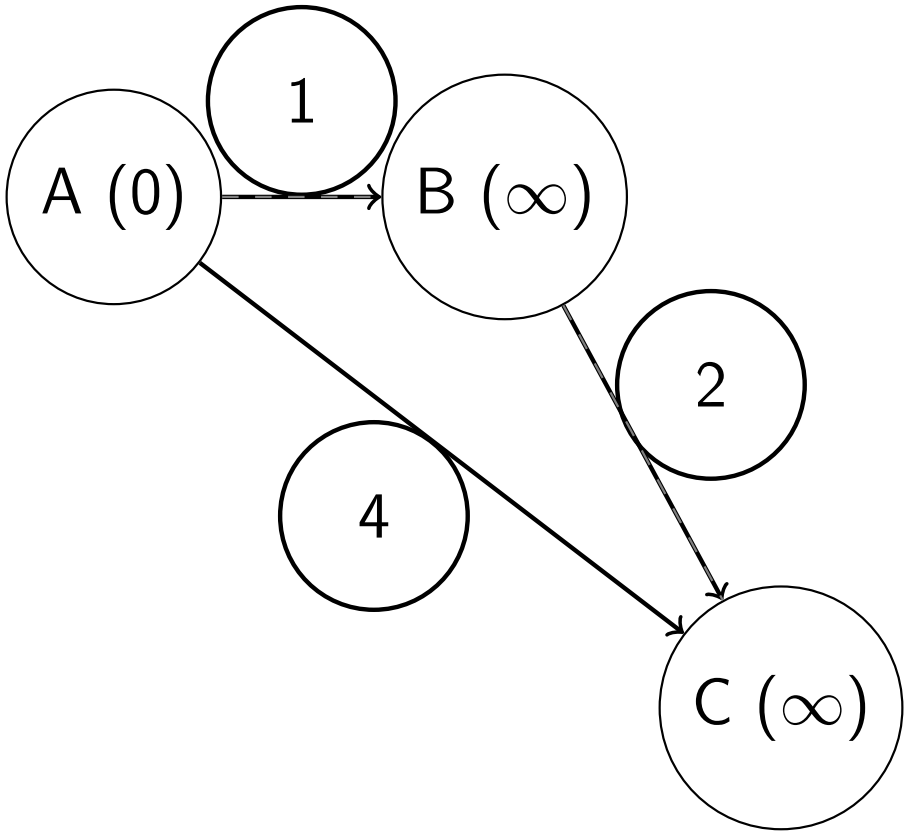
Conclusion

- The LP formulation models the shortest path as a min-cost flow problem with unit supply/demand.
- Advantages: Handles variants (e.g., capacities); solvable with LP solvers.
- Note: For pure shortest path, integer solutions are guaranteed due to network structure.

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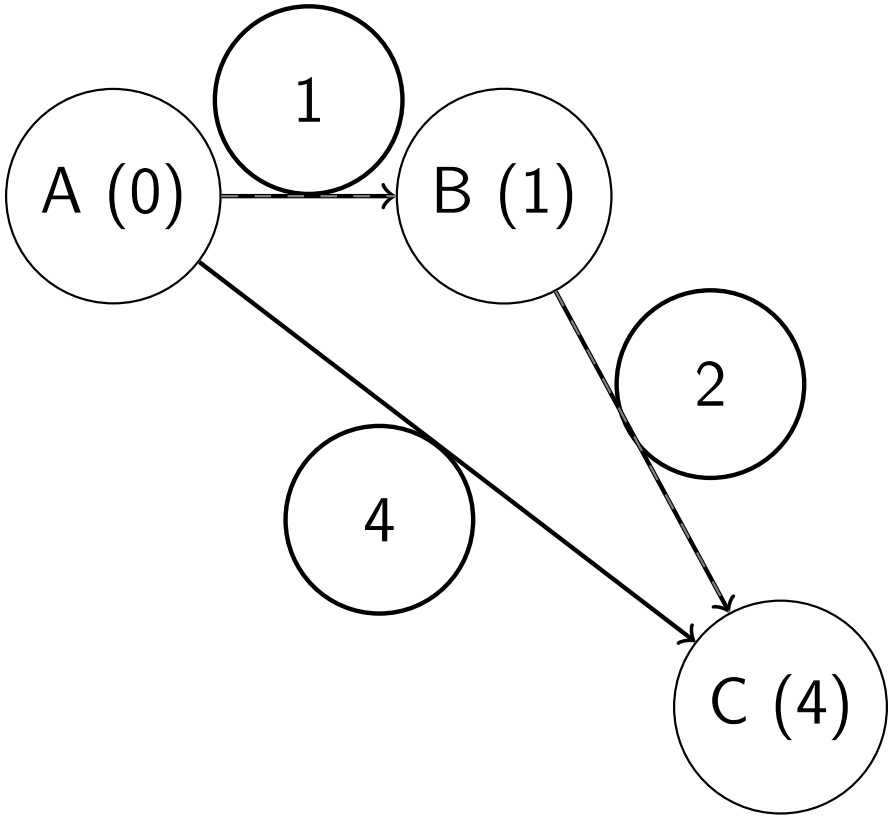
Dijkstra's Algorithm: Step 1 - Initialization

Find the shortest path from A to C.



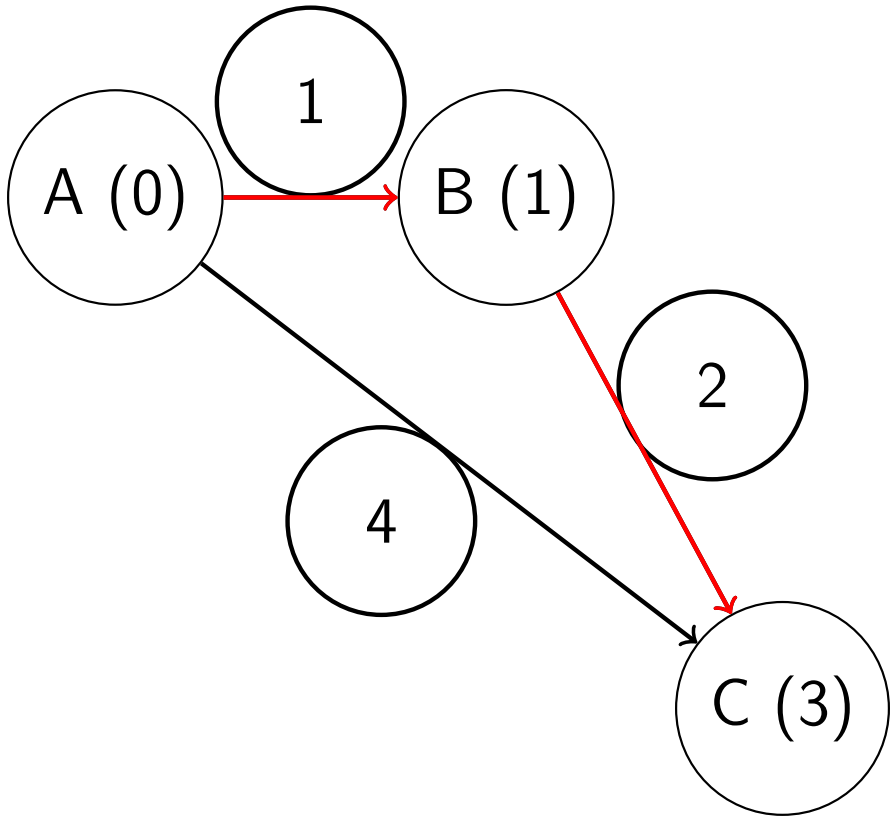
- Start at node A, set distance to 0. - Initialize distances to all other nodes as ∞. - Set A as the current node.

Dijkstra's Algorithm: Step 2 - Update Distances



- Update distances from A: - B: $0 + 1 = 1$ (update from ∞) - C: $0 + 4 = 4$ (update from ∞) - Mark A as visited. - Move to unvisited node with smallest distance: B (1).

Dijkstra's Algorithm: Step 3 - Final Path



- Update distances from B: - C: $\min(4, 1+2) = 3$ (update from 4) - Mark B as visited. - Only unvisited node is C (3), mark as visited. - Shortest path: A $\rightarrow$ B $\rightarrow$ C with total cost 3.

Sensitivity Analysis in Linear Programming

Shadow Prices and Reduced Costs

- Using Par, Inc. Golf Bag Production as Example

Introduction to Sensitivity Analysis

Sensitivity analysis examines how changes in the LP parameters (coefficients in objective, RHS of constraints, etc.) affect the optimal solution.

Why important?

- Helps understand the robustness of the solution.
- Provides managerial insights (e.g., value of resources).
- Identifies critical parameters.

Key concepts:

- Shadow prices (for constraints)
- Reduced costs (for variables)
- Ranges of validity

Recap of Par, Inc. Problem

Decision variables:

Let x_s = number of standard golf bags produced per week,
x_d = number of deluxe golf bags produced per week.
Both $x_s, x_d \geq 0$.

Objective function:

Maximize weekly profit:

$$\max \ Z = 10x_s + 9x_d$$

Subject to capacity constraints:

$$\begin{aligned}
\frac{7}{10}x_s + x_d &\leq 630 && \text{(Cutting \& Dyeing)} \\
\frac{1}{2}x_s + \frac{5}{6}x_d &\leq 600 && \text{(Sewing)} \\
x_s + \frac{2}{3}x_d &\leq 708 && \text{(Finishing)} \\
0.1x_s + 0.25x_d &\leq 135 && \text{(Inspection \& Packaging)} \\
x_s, x_d &\geq 0
\end{aligned}$$

Writing the PuLP Code

ChatGPT Prompt

Here is an linear program: max $Z = 10x_s + 9x_d$ s.t. $\frac{7}{10}x_s + x_d \leq 630; \frac{1}{2}x_s + \frac{5}{6}x_d; x_s + \frac{2}{3}x_d \leq 708; 0.1x_s + 0.25x_d \leq 135; x_s, x_d \geq 0$. Write me a python code that solves the linear program using PuLP. Also add code that will compute **reduced costs**, **shadow prices**, and **indicate which constraints are loose or tight**.

PuLP Code with Shadow Prices and Reduced Costs

```
import pulp as pl

# Model
prob = pl.LpProblem("Par_Inc_Golf_Bags", pl.LpMaximize)

# Variables (continuous  0; add cat='Integer' if you ever need integrality)
x_s = prob.add_variable("Standard_Bags", lowBound=0)
x_d = prob.add_variable("Deluxe_Bags",   lowBound=0)

# Objective
prob += 10 * x_s + 9 * x_d, "Maximize_Profit"

# Constraints
prob += 0.7 * x_s + 1.0   * x_d <= 630, "Cutting_Dyeing"
prob += 0.5 * x_s + (5/6) * x_d <= 600, "Sewing"
prob += 1.0 * x_s + (2/3) * x_d <= 708, "Finishing"
prob += 0.1 * x_s + 0.25  * x_d <= 135, "Inspection_Packaging"

# Solve with CBC so reduced costs are available
res = prob.solve()
```

PuLP Code with Shadow Prices and Reduced Costs

```
# Report primal solution
print("Status:", res.status.name)
print("Optimal Solution:")
print(f"  Standard Bags = {x_s.varValue:.6f}")
print(f"  Deluxe Bags   = {x_d.varValue:.6f}")
print(f"  Maximum Profit = ${pl.value(prob.objective):.6f}")

# Reduced costs for variables (dj):
# For a maximization, a nonbasic variable has dj <= 0.
print("\nReduced Costs (variables.dj):")
for v in prob.variables():
    dj_str = f"{v.dj:.6f}" if v.dj is not None else "None"
    print(f"  {v.name:20s}  dj = {dj_str}")

# Shadow prices (constraint.pi) and slacks (constraint.slack)
# For a <= constraint, slack = RHS - LHS at the optimum.
print("\nShadow Prices and Slacks:")
TOL = 1e-7  # numerical tolerance for binding check
for c in prob.constraints():
    cname = c.name
    dual = c.pi
    slack = c.slack
    # Classify binding if slack ~ 0
    is_binding = abs(slack) <= TOL
    print(f"  {cname:22s}  dual = {dual: .6f}   slack = {slack: .6f}   binding = {
        is_binding}")

# If you also want a compact "which constraints bind" list:
binding_constraints = [name for name, c in prob.constraints.items() if abs(c.slack) <=
    TOL]
print("\nBinding constraints:", binding_constraints if binding_constraints else "None")
```

Optimal Solution

No need to read the code. Just run it to get this:

Optimal Production Plan

- **Standard Bags** (x_s): 540
- **Deluxe Bags** (x_d): 252

Maximum Profit

$$\$7,668.00$$

Binding Constraints

These resources are fully utilized (Slack $= 0$) and define the optimal solution.

- **Cutting/Dyeing** ($\mathbf{C_1}$): $0.7x_s + 1.0x_d \leq 630$
- **Finishing** ($\mathbf{C_3}$): $1.0x_s + \frac{2}{3}x_d \leq 708$

Shadow Prices

Shadow Price Definition

Shadow price (or dual price) for a constraint is the rate of change in the optimal objective value with respect to a unit change in the right-hand side (RHS) of that constraint.

Shadow Prices (π_i): Marginal Value of Resources

Resource	**Current RHS**	**Shadow Price (π_i)**	**Slack**
Cutting/Dyeing (C_1)	630 hours	**\$4.3750**	0.00
Sewing (C_2)	600 hours	**\$0.0000**	120.00
Finishing (C_3)	708 hours	**\$6.9375**	0.00
Inspection/Pkg. (C_4)	135 hours	**\$0.0000**	18.00

Interpretation: An extra hour in the **Finishing** department is worth \$6.94 in marginal profit.

Reduced Costs

Reduced Cost Definition

Reduced cost is the difference between its marginal contribution to the OBJ function value and the marginal value of the resources it consumes.

Reduced Costs

Variable	Current c_j	Reduced Cost
$\mathbf{x_s}$ (Standard)	\$10.00	0.00
$\mathbf{x_d}$ (Deluxe)	\$9.00	0.00

Spreadsheet-Based Sensitivity Report

Sensitivity Report

Adjustable Cells

Cell	Name	Final Value	Reduced Cost	Objective Coefficient	Allowable Increase	Allowable Decrease
B4	Standard Bag (x1)	540	0	10	3.5	3.7
C4	Deluxe Bag (x2)	252	0	9	5.285714286	2.333333333

Constraints

Cell	Name	Final Value	Shadow Price	Constraint R.H. Side	Allowable Increase	Allowable Decrease
D9	Cutting & Dyeing LHS	630	4.375	630	52.36363636	134.4
D10	Sewing LHS	480	0	600	1E+30	120
D11	Finishing LHS	708	6.9375	708	192	128
D12	Inspection & Packaging LHS	117	0	135	1E+30	18

Sensitivity Ranges *without* Using Excel

ChatGPT Prompt

Here is an linear program: max $Z = 10x_s + 9x_d$ s.t.
$\frac{7}{10}x_s + x_d \leq 630; \frac{1}{2}x_s + \frac{5}{6}x_d; x_s + \frac{2}{3}x_d \leq 708; 0.1x_s + 0.25x_d \leq 135; x_s, x_d \geq 0.$
Give me **reduced costs**, **shadow prices**, and **indicate which constraints are loose or tight** and **sensitivity ranges**.

Sensitivity Ranges for Reduced Costs

Range of Optimality (Profit Coeffs)

The optimal plan remains $x_s = 540, x_d = 252$ as long as the profit coefficients (c_j) stay within these ranges.

Variable	**Current c_j**	**Valid Range**
$\mathbf{x_s}$ (Standard)	\$10.00	[\$6.50, \$13.50]
$\mathbf{x_d}$ (Deluxe)	\$9.00	[\$6.67, \$14.29]

Calculation of Valid Range:

- Some software reports "*allowable increase*" and "*allowable decrease*" instead of "*valid range*"
- 13.70=10+allowable increase=10+3.5-13.5
- 6.3=10-allowable decrease=10-3.5

Sensitivity Ranges

Range of Feasibility (RHS Values)

The Shadow Prices remain constant as long as the resource limits (b_i) stay within these ranges.

Resource	**Current** b_i	**Valid Range**
Cutting/Dyeing	630	$[495.60, 682.36]$
Finishing	708	$[580, 900.00]$

Note: Non-binding constraints have an infinite upward range.

Resource Value: Impact of Increased Finishing Hours

Analysis Goal

Question: What happens to the Maximum Profit if we increase the Finishing capacity (C_3) by 2 hours?

1. **Identify Key Metric:** The shadow price (π) for the Finishing constraint is $\$\mathbf{6.9375}$.
2. **Check Validity Range:** The proposed new RHS is $708 + 2 = 710$ hours. This is within the valid range of $[580, 900]$, so the shadow price remains valid.
3. **Calculate Profit Change (ΔZ):**

$$\begin{aligned}\Delta Z &= \pi_{\text{Finishing}} \times \text{Extra Hours} \\ &= \$6.9375 \times 2 \\ &= \mathbf{\$13.875}\end{aligned}$$

4. **Calculate New Maximum Profit (Z'):**

$$\$\mathbf{7,668.00} + \$\mathbf{13.88} = \$\mathbf{7,681.88}$$

Conclusion: Investing in 2 extra Finishing hours will increase the total maximum profit by \$13.88.

Another Application of Shadow Prices

Suppose Par, Inc. can purchase additional labor hours:

- For Cutting & Dyeing at \$4/hour: Since shadow price \$4.375 $>$ \$4, profitable to buy.
- For Finishing at \$7/hour: \$6.9375 $<$ \$7, not profitable.
- For Sewing at any positive cost: Not worth it (shadow price 0).

This helps in decision-making for resource allocation.

Reduced Cost of Basic Variable (x_s)

Computation of Reduced Cost

Opportunity Cost is the total value of the scarce resources consumed by one unit of the product.

- **Marginal Profit** (c_s for Standard Bag): \$10.00
- **Opportunity Cost**: $\sum$(Resource Usage $\times$ Shadow Price)

$$d_s = \text{Profit} - \text{Opportunity Cost}$$

$$d_s = \underbrace{10.00}_{\text{marginal profit}} - \Big(\underbrace{4.3750}_{\substack{\text{shadow price} \\ \text{of C\&D}}} \times \underbrace{0.7}_{\substack{\text{hours} \\ \text{of C\&D} \\ \text{needed}}} + \underbrace{6.9375}_{\substack{\text{shadow price} \\ \text{of finishing}}} \times \underbrace{1.0}_{\substack{\text{hours} \\ \text{of finishing} \\ \text{needed}}} \Big)$$

$$d_s = 10.00 - (3.0625 + 6.9375) = 0.00$$

Result: The per-unit profit of \$10 for a Standard Bag is exactly balanced by the \$10 worth of scarce resources (C&D and Finishing) it consumes.

Practicing Sensitivity Analysis in Linear Programming

Parket Sisters: Pencil and Pen Factory

- Read the case study posted on Canvas

Parket Sisters

We model Parket Sisters' weekly production planning as a linear program. The decision variables are the weekly production quantities of three products:

$$X_1 = \#\text{ of ballpoint pens},$$
$$X_2 = \#\text{ of mechanical pencils},$$
$$X_3 = \#\text{ of fountain pens}.$$

The objective maximizes weekly net profit (in dollars):

$$\max 3.0X_1 + 3.0X_2 + 5.0X_3.$$

Subject to material capacity constraints (ounces available per week):

$$\begin{aligned} \text{Plastic:} \quad & 1.2X_1 + 1.7X_2 + 1.2X_3 \leq 1{,}000 \\ \text{Chrome:} \quad & 0.8X_1 + 0 \cdot X_2 + 2.3X_3 \leq 1{,}200 \\ \text{Stainless steel:} \quad & 2.0X_1 + 3.0X_2 + 4.5X_3 \leq 2{,}000 \\ & X_1, X_2, X_3 \geq 0. \end{aligned}$$

Case Questions: Baseline Solution

Weekly optimal mix? Weekly optimal net profit?

Solution reported:

$$X_1 = 700,\ X_2 = 0,\ X_3 = 133.3\overline{3},\ \text{Profit} = 2{,}766.67.$$

Sensitivity Report

Adjustable Cells

Cell	Name	Final Value	Reduced Cost	Objective Coefficient	Allowable Increase	Allowable Decrease
C4	Ballpoint pens (X1)	700	0	3	2	0.777778
D4	Mechanical pencils (X2)	0	-1.38333	3	1.38333	1.00E+30
E4	Fountain pens (X3)	133.3333333	0	5	1.75	2

Constraints

Cell	Name	Final Value	Shadow Price	Constraint R.H. Side	Allowable Increase	Allowable Decrease
F9	Plastic LHS	1000	1.16667	1000	200	466.667
F10	Chrome LHS	866.6666667	0	1200	1.00E+30	333.333
F11	Stainless Steel LHS	2000	0.8	2000	555.556	333.333

Case Question 4: Shadow Prices

What is the marginal value of one more unit of chrome? Of plastic?

From the sensitivity report:

- *Shadow price of chrome* $= 0$;
- *shadow price of plastic* $= 1.167$.

Case Question 5: Extra Stainless Steel Offer

Should Parket Sisters accept an additional 500 oz of stainless steel at \$0.60 per oz above the usual price?

Case Question 5: Extra Stainless Steel Offer

Should Parket Sisters accept an additional 500 oz of stainless steel at \$0.60 per oz above the usual price?

Because $500 < 555.56$ *(allowable increase for stainless steel), the current shadow price applies. With shadow price* \$0.80 *and incremental cost* \$0.60, *the net contribution is positive.* **Accept the offer.**

Case Question 10: Reduced Chrome Supply

What if the supplier only sends 1,000 oz of chrome.

Case Question 10: Reduced Chrome Supply

What if the supplier only sends 1,000 oz of chrome.

If the chrome supplier can send only 1,000 *oz (instead of* 1,200*), no change occurs because the baseline solution uses only* 866.67 *oz, leaving* 333.33 *oz slack.* **No effect on mix or profit.**

Case Question 11: New Mechanical Pencil Design

New Product Profile

The new Mechanical Pencil design has a unit **Profit (c_j)** of $\$\mathbf{3.00}$.

Resource Consumption per unit:

- Plastic: 1.1 oz; Chrome: 2.0 oz; Stainless Steel: 2.0 oz

Resource Shadow Prices (π_i)

The marginal value of resources for the current optimal plan are assumed to be:

- Plastic (π_1): \$1.167; Chrome (π_2): \$0.00; Stainless Steel (π_3): \$0.80

Reduced Cost Calculation Goal

We compute the Reduced Cost (d_j) to determine if this new product is profitable enough to be produced:

$$d_j = \text{Profit} - \text{Opportunity Cost (Resource Value)}$$

Case Question 11: Calculation & Decision

Implied Resource Opportunity Cost

This is the total cost of the valuable resources consumed by one new pencil.

$$\begin{aligned}\text{Opportunity Cost} &= (1.1)(1.167) + (2.0)(0) + (2.0)(0.80) \\ &= 1.2837 + 0 + 1.60 \\ &\approx \mathbf{\$2.88}\end{aligned}$$

Reduced Cost (Marginal Profit Net of Cost)

The reduced cost determines the net value of the new product at the current optimal basis.

$$\begin{aligned}d_j &= \text{Unit Profit} - \text{Opportunity Cost} \\ &= 3.00 - 2.8837 \\ &\approx \mathbf{\$0.12}\end{aligned}$$

DECISION

Since $\mathbf{d_j = \$0.12 > 0}$, **approve the design**. Re-solve the LP to find the new optimal production mix.